

TreEAID (Treefall Evaluation, Analysis, and Identification through Deep Learning) User Guide

Mitra Nasimi
Richard L. Wood*
Corresponding author: rwood@unl.edu

Department of Civil and Environmental Engineering
University of Nebraska, Lincoln
Lincoln, Nebraska

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The software outlined in this manual is provided "as is," without any warranties, expressed or implied. Users acknowledge and assume all risks associated with its use. The developer is not liable for any errors or damages, direct or indirect, arising from the use of the software. There is no guarantee of error-free or uninterrupted operation, and the developer reserves the right to make changes without notice. Users are responsible for confirming and validating the results. By using the software, users agree to these terms.

2 Nomenclature and Symbols

Symbol	Description
°	Degree
A	Area of Best-Fit Ellipse (Root Ball)
a	Major Axis of the Ellipse (Root Ball)
α	Alpha Value Used for Post-Processing Detection Masks
API	Application Programming Interface
AR	Aspect Ratio
b	Minor Axis of the Ellipse (Root Ball)
CL	Confidence Level
CRS	Coordinate Reference System
CSV	Comma-Separated Values
DEM	Digital Elevation Model
EPSG	European Petroleum Survey Group (Used to Define Coordinate Reference Systems)
exe	Windows Executable File Extension
GIS	Geographic Information System
GSD	Ground Sampling Distance
GUI	Graphical User Interface
HAS	Horizontal Angle Sign
IoU	Intersection over Union
JPG	Joint Photographic Experts Group Image Format
JSON	JavaScript Object Notation
L	Tree Length
$L_{\text{extension}}$	Extension Length Extruded from Each End of the Tree Polygon
m	SI Unit of Meter
N	Tree Thickness Multiplier
NMS	Non-Maximum Suppression
PGW	World File Associated with the Images in PNG Format for Spatial Referencing
PIE	Percent of Interior Extension
PNG	Portable Network Graphics Image Format
pt	PyTorch Model Weight File Extension
px	Pixels
QGIS	Open-Source Geographic Information System Software
rad	Radian
r	Pearson Correlation Coefficient
ROI	Region of Interest
SPC	State Plane Coordinate System
t	Tree Thickness
θ_{final}	Final Aggregated Orientation Angle of the Treefall Group
θ_{href}	Reference Horizontal Angle (Used for Orientation Comparison)
TIF/ TIFF	Tagged Image File Format
TreEAID	Treefall Evaluation, Analysis, and Identification through Deep Learning
TR	Taper Rate
UAS	Uncrewed (Unpiloted) Aerial System
WGS	World Geodetic System

XML	Extensible Markup Language
YOLO	You Only Look Once

3 Introduction

TreEAID (Treefall Evaluation, Analysis, and Identification through Deep Learning) is a Windows-based software application designed for the automated analysis of high-resolution ortho imagery. It enables users to:

- 1) Detect tornado-induced treefall (fallen tree trunks)
- 2) Identify the fall direction of each tree instance
- 3) Perform spatial subsampling of the treefall directions to create a wind direction map
- 4) Detect the root ball (or root plate) of the uprooted trees (upon user request)
- 5) Identify the thickness and the width of the root ball instances
- 6) Perform spatial subsampling to provide the number of uprooted trees and the average root ball area of the region

The application is primarily designed for post-tornado imagery in forested regions, but it is applicable to other windstorm-related treefall scenarios. It can output geotagged treefall vectors and annotated images (upon user selection), as detailed below. Sample treefall outputs are illustrated in Figure 1 and Table 1, Table 2. TreEAID can significantly expedite the manual tagging process for treefall directions following tornadoes or other windstorm events.

- 1) Treefall directions (horizontal and azimuth angles)
- 2) Treefall vector start and end points in both the State Plane Coordinate System (SPCs) and the World Geodetic System (WGS)
- 3) Confidence levels, thickness, and length of fallen trees in pixels and global coordinate system (e.g., meters (m))
- 4) Taper rates for each detected treefall instance [1].

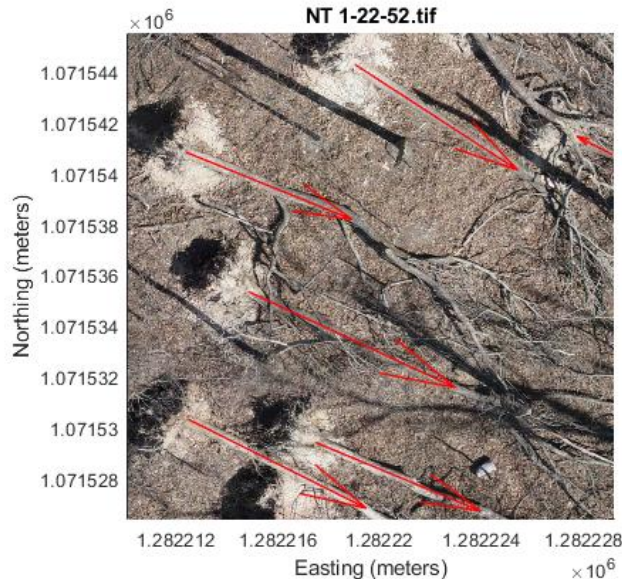


Figure 1. Annotated output with tagged treefall angle instances.

Table 1. Application output showing the start and end points of the fallen tree vector in State Plane Coordinate System (SPCs) and the World Geodetic System 1984 (WGS84).

X Start [m]	Y Start [m]	X End [m]	Y End [m]	Lon Start [°]	Lat Start [°]	Lon End [°]	Lat End [°]
1282350.9	1071465.8	1282359.6	1071460.8	-88.1938	36.9516	-88.1937	36.9515
1282357.9	1071463.9	1282353.3	1071468.8	-88.1937	36.9516	-88.1938	36.9516
1282345.6	1071450.4	1282344.3	1071453.9	-88.1939	36.9514	-88.1939	36.9515
1282356.8	1071463.3	1282350.3	1071466.8	-88.1938	36.9515	-88.1938	36.9516
1282360.3	1071452.1	1282362.5	1071453.5	-88.1937	36.9514	-88.1937	36.9515
1282363.3	1071463.3	1282361.5	1071464.5	-88.1937	36.9515	-88.1937	36.9516

Table 2. Application output showing the fall angles and geometric attributes for each detected treefall.

Azimuth Angle (Clockwise from North) [°]	Horizontal Angle [°]	Confidence Level [unitless]	Tree Thickness [px]	Tree Thickness [m]	Tree Length [px]	Tree Length [m]	Taper Rate [m/m]
119.82	330.18	0.90	21.21	0.40	540.67	10.12	-0.01
316.95	133.05	0.88	20.44	0.38	367.72	6.88	0.01
338.44	111.56	0.82	17.57	0.33	226.37	4.24	0.03
298.53	151.47	0.80	14.48	0.27	401.84	7.52	0.01
57.72	32.28	0.79	7.15	0.13	145.65	2.72	-0.01
304.59	145.41	0.69	13.90	0.26	123.31	2.31	0.00
304.88	145.12	0.58	14.33	0.27	131.54	2.46	0.04

Upon user request, the application can also identify uprooted root balls and extract their locations, dimensions, and orientations. Outputs include the center locations (in SPCs and WGS), thickness and width, and rotational angles (in radians, horizontal degrees, and azimuth degrees). Thickness and width values are either in a defined global coordinate system (e.g., meters for inputs in GeoTIFF or PNG formats) or in pixel units (for JPG images). The root ball results are in Figure 2 and Table 3, Table 4. These data can offer important insight into the ecological impacts of tornadoes on forested environments.

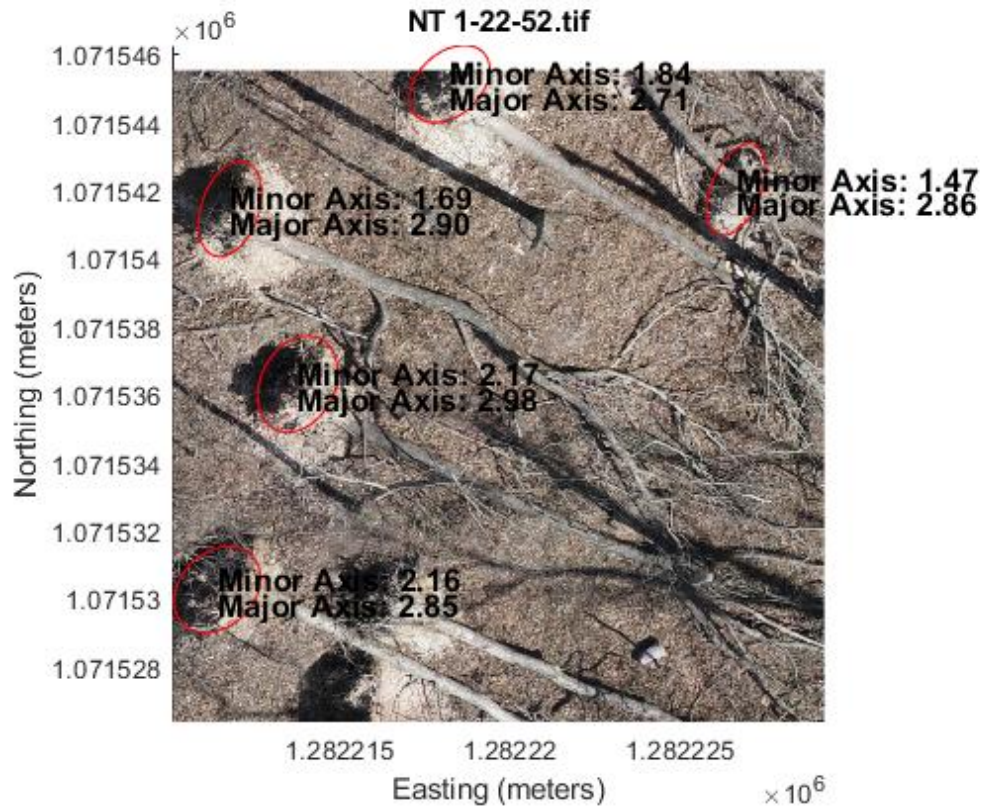


Figure 2. Annotated output with tagged root ball instances and associated dimensions.

Table 3. Application output for the root ball evaluation showing the center of the detected root balls in the State Plane Coordinate System (SPCs), the World Geodetic System 1984 (WGS84), and the confidence level of the model detection.

X Center [m]	Y Center [m]	Lon Center [°]	Lat Center [°]	Confidence Level [Unitless]
1282358.71	1071463.72	-88.1937	36.9516	0.79
1282348.92	1071463.36	-88.1938	36.9515	0.17
1282378.89	1071460.66	-88.1935	36.9515	0.78
1282364.02	1071461.81	-88.1937	36.9515	0.58
1282397.58	1071466.03	-88.1933	36.9516	0.84
1282390.39	1071467.51	-88.1934	36.9516	0.82

Table 4. Application output for the root ball evaluation showing the orientation and geometric attributes for each detected root ball.

Major Axis Length [m]	Minor Axis Length [m]	Rotation Angle [rad]	Rotation Angle Horizontal [°]	Rotation Angle Azimuth [°]
1.72	1.25	8.61	223.30	226.70
1.24	0.59	-9.45	268.68	181.32
2.37	1.50	-9.86	244.86	205.14
1.43	0.48	8.96	243.46	206.54
3.35	1.78	-9.95	239.72	210.28
2.97	1.49	-9.91	242.05	207.95
2.63	1.30	-9.76	251.05	198.95

TreEAID (version 4.1.0) supports both local and cloud-based model inference. The cloud-based model is limited to 50,000 images per month across all users and requires a stable internet connection, but it is useful for systems with limited processing power. The local model is preferred and recommended for large-scale processing, as it operates without the need for internet connectivity. In both modes, Python and associated packages must be installed. The software is available only for Windows and is not officially supported on Mac or Linux operating systems.

4 Input Requirements

TreEAID is built on a deep learning model trained with tiled orthophotos and can process images of various sizes. For optimal performance, input images should be 1024×1024 pixels, as deep learning models are sensitive to scale and input resolution. Tiles should be non-overlapping in both northing and easting directions. Software such as QGIS or ArcGIS can be used to divide orthomosaics into uniform tiles. Figure 3 presents aerial imagery samples with varying ground sampling distances (GSD): 2 cm, 5 cm, and 10 cm, each with a size of 1024×1024 pixels. The presence of visible fallen trees and root balls in each image is critical for successful detection. The application has been tested on images with GSDs ranging from approximately 1.5 to 10 cm.

TreEAID predicts the treefall orientations and root ball dimensions when inputs meet the specified criteria. The software accepts PNG, TIF, and JPG image formats. Orthomosaic images must be undistorted and nadir-oriented to ensure prediction accuracy. This requirement is satisfied by using orthomosaic images of surveyed regions that are tiled without overlapping. These are typically generated from aerial surveys and georeferenced using photogrammetry (e.g., Agisoft MetaShape) or GIS platforms (e.g., QGIS and ArcGIS).

1. For TIF (GeoTIFF) images, geographic metadata, including the coordinate reference system (CRS) and EPSG code is embedded directly within the file. This enables automatic geotagging of outputs in real-world coordinates.
2. For PNG images, a world file (PGW) and the corresponding EPSG code must be provided to enable geospatial processing.
3. For JPG images, which lack embedded or linked geospatial metadata, all results will be output in pixel coordinates only. These are intended primarily for testing and not for quantitative spatial analysis.

Users should ensure that each image has consistent projection information. EPSG codes can be verified on GIS platforms or online resources such as EPSG.io [2]. It is also recommended to organize input images and any auxiliary files (e.g., world files) within a single folder prior to analysis. TreEAID saves intermediate and final outputs in subfolders within the input directory, including detection masks, georeferenced treefall vectors, root ball metrics, and annotated images (if required). Users should ensure adequate disk space is available to store all output files.

To summarize the image format capabilities:

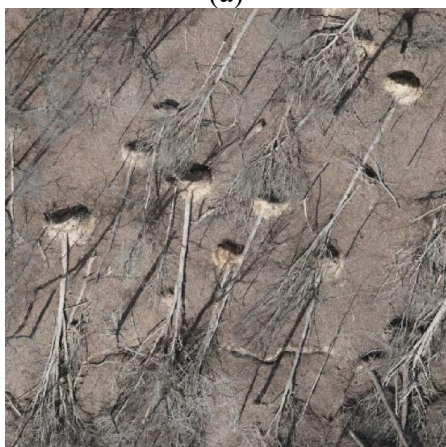
- 1) GEOTIFF (*.tif): Full geospatial support and this is the preferred format
- 2) PNG: Geospatially supported when .pgw and the EPSG code are provided
- 3) JPG: No geospatial metadata and the outputs are limited to the pixel coordinates. This is not suitable for spatial analysis.



(a)



(b)



(c)



(d)



(e)



(f)

Figure 3. Input images at ground sampling distance (GSD): (a, b) 2 cm; (c, d) 5 cm; and (e, f) 10 cm. All images are 1024 x 1024 pixels.

5 Required Software and Packages

TreEAID is developed as a hybrid software system combining MATLAB and Python, leveraging the graphical user interface (GUI) and runtime stability of MATLAB alongside the deep learning and computer vision capabilities of Python. The application is compiled as a standalone Windows executable and runs using the MATLAB Runtime. While most dependencies are embedded, a few Python packages require separate installation in specific versions. Users should review the installation instructions carefully before using the software. This section reviews all required packages and explains the step-by-step installation process.

5.1 Required Software

Python: The application requires Python version 3.10.11, which can be downloaded at <https://www.python.org/downloads/release/python-31011/>. To install the required packages, open the Command Prompt by typing “cmd” or “Command Prompt” into the Windows search bar.

5.2 Required Packages

The following Python packages must be installed manually using the pip command.

- Ultralytics – Required for image segmentation, object detection, and other core model functions. For compatibility, install version 8.3.58:

```
pip install ultralytics==8.3.58
```

- OpenCV-Python – Used for object identification and image processing:

```
pip install opencv-python
```

- Alphashape – Used to reconstruct complex shapes from coordinate points (e.g., treefall fragments):

```
pip install alphashape
```

- Roboflow – Required only for cloud-based functionality. Roboflow supports dataset management, model inference, and remote weight access:

```
pip install roboflow
```

6 Installing Application

The application is distributed as a unified package with a .exe extension. The installer file is named:

- TreEAID 4.1.0.exe

During the installation, the required MATLAB runtime will be automatically downloaded. This process may take several minutes, depending on the internet speed. This application is compatible only with Windows and is not supported on Mac or Linux operating systems. Figure 4 displays the application's installation window. After installation, the application can be launched by double-clicking the desktop icon or searching for it in the Start menu.

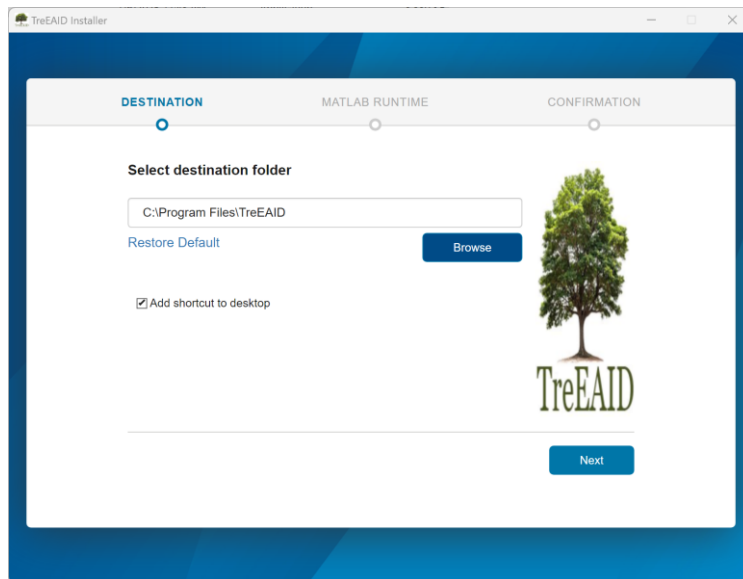


Figure 4. Software installation window during setup.

7 Input and Treefall Results Tab

The Input and Treefall Results tab contains the main processing section of the application, and it is the first tab of the application. In this tab, the application requires different inputs for the local and cloud models to start processing. Therefore, some input fields may be deactivated depending on the selected deep learning model. Before starting the processing, users must organize the input dataset in a folder. It is important to note that the application will process and save the results of each step in subfolders within the designated input folder, as separate folders for both treefall and root ball results. To initiate processing, users must follow the steps detailed below.

7.1 Image and Coordinate Format

This section defines the path to the input image folders, the image format, and, more importantly, the EPSG code (or projection code). Users must first select the folder containing the input images, as shown in Figure 5.

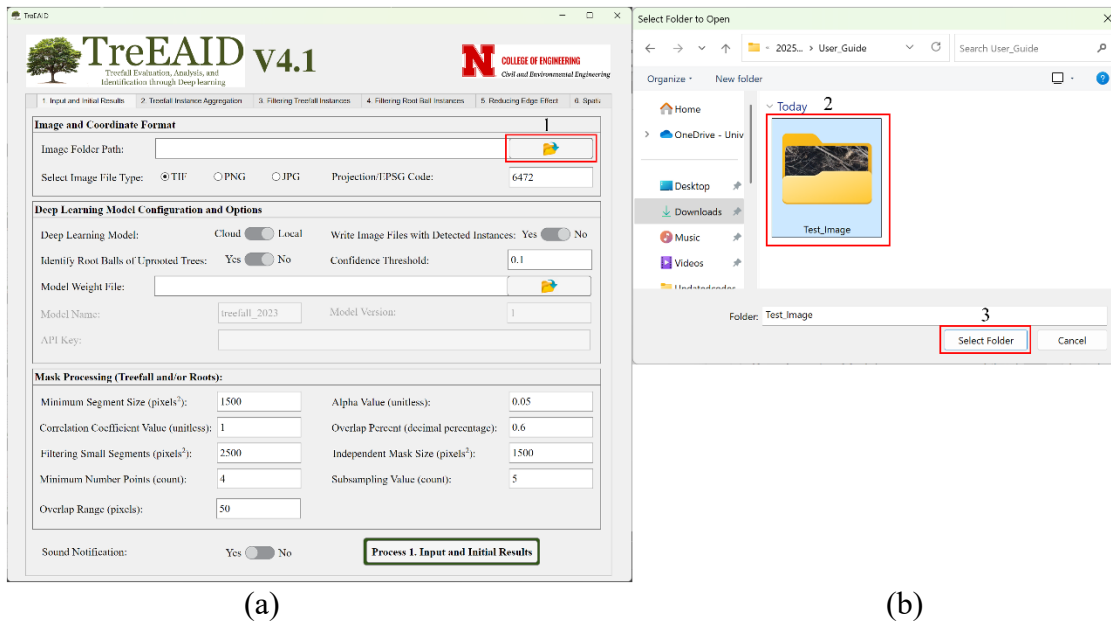


Figure 5. Image folder selection: (a) redirecting the user to select the image folder; (b) selecting the image folder location.

After selecting the image folder, users must specify the image type/format. The application supports images in TIF, PNG, and JPG formats. Figure 6 displays the section for identifying the file type of the imagery.

- **GeoTIF:** Geographic information must be embedded in the TIF imagery dataset. These datasets can be analyzed with or without associated XML files, and the application can generate geotagged results. This is the preferred input format for the application to prevent any complications that may occur during processing due to the selection of an incorrect EPSG code.
- **PNG:** For processing, PNG files must be accompanied by a PGW (world) file to enable geospatial processing. Images in this format can produce geotagged results.

- **JPG:** Although supported, JPG images lack embedded geospatial information, limiting output to pixel coordinates. This format is primarily supported for testing purposes, as the results are not geotagged.

The screenshot shows the TreEAID v4.1 software interface. The 'Image and Coordinate Format' section is highlighted, showing the 'Select Image File Type' section with radio buttons for TIF, PNG, and JPG. The 'Projection/EPSG Code' field is set to 6472. The 'Deep Learning Model Configuration and Options' section includes settings for the model, file paths, and confidence threshold. The 'Mask Processing (Treefall and/or Roots)' section includes settings for segment size, correlation coefficient, and overlap range. A 'Process 1: Input and Initial Results' button is visible at the bottom right.

Figure 6. Selection of the supported image format.

After specifying the image format, the user must enter the EPSG code (coordinate reference system or projection). The EPSG code defines the coordinate reference system (CRS) used to georeference the orthoimages. Identifying the correct EPSG code for the surveyed region is essential. The model utilizes this code to define the coordinate reference system (CRS), which encompasses the datum and projection parameters necessary for accurately georeferencing each image in world coordinates. It typically consists of a 4 to 5 digit number and provides details about the coordinate reference system (CRS), such as map projection, datum, and the units of the data.

Inputs in GeoTIF format do not require an EPSG code at this stage of processing, because the coordinate reference system (CRS) and the origin coordinate of each orthophoto are typically embedded in the file's metadata. The EPSG code is only needed later, during the subsampling step, for GeoTIF images. This is because subsampling uses only the georeferenced output (CSV file), and a correct EPSG code allows for proper conversion between the State Plane Coordinate System and the World Geodetic System accurately.

For the inputs in PNG format, it is essential to ensure that the images and the provided EPSG code are compatible with each other, since PNG files do not contain embedded geospatial metadata. Georeferencing for the PNG images is performed using the accompanying world file (PGW), which provides the image scale and origin. To correctly map the computed values from the image coordinate system (pixels) to real-world coordinates, the application scales the output values from pixels using the values extracted from the world file. The origin coordinates define the image position in space, so they are used to transfer the computed features to image space. The datum is then used to align the computed values correctly with their real-world coordinate.

Selecting the correct EPSG code is essential for accurate spatial analysis, especially when integrating outputs with other geospatial datasets, such as digital elevation models (DEMs), digital terrain models (DTMs), or other raster and vector datasets. Users can verify geographic metadata by opening the image in GIS software and reviewing its properties.

Figure 7 shows the section where the EPSG code needs to be entered. For example, the default EPSG code of the application is 6472, which corresponds to "Kentucky: EPSG 6472 - NAD83 (2011)-Kentucky Single Zone" and is selected as an example. Users can determine the appropriate EPSG code using GIS software platforms, such as QGIS, ArcGIS, or EPSG.io, which provides details about coordinate systems worldwide [2].

The screenshot displays the TreEAID v4.1 software interface. At the top, there is a header with the application logo and name, and a navigation bar with six tabs: 1. Input and Initial Results, 2. Treefall Instance Aggregation, 3. Filtering Treefall Instances, 4. Filtering Root Ball Instances, 5. Reducing Edge Effect, and 6. Spots. The '1. Input and Initial Results' tab is active. Below the header, the 'Image and Coordinate Format' section contains fields for 'Image Folder Path' (C:\Users\mhzam\OneDrive\Desktop\TEST_IMAGES\TIF_TEST), 'Select Image File Type' (radio buttons for TIF, PNG, and JPG), and 'Projection/EPSG Code' (a text box containing '6472', which is highlighted with a red rectangle). Below this, the 'Deep Learning Model Configuration and Options' section includes settings for 'Deep Learning Model' (Cloud/Local), 'Write Image Files with Detected Instances' (Yes/No), 'Identify Root Balls of Uprooted Trees' (Yes/No), 'Confidence Threshold' (0.1), 'Model Weight File' (treefall_2023), 'Model Version' (1), and 'API Key'. The 'Mask Processing (Treefall and/or Roots)' section contains various numerical inputs: 'Minimum Segment Size (pixels²)' (1500), 'Alpha Value (unitless)' (0.05), 'Correlation Coefficient Value (unitless)' (1), 'Overlap Percent (decimal percentage)' (0.6), 'Filtering Small Segments (pixels²)' (2500), 'Independent Mask Size (pixels²)' (1500), 'Minimum Number Points (count)' (4), 'Subsampling Value (count)' (5), and 'Overlap Range (pixels)' (50). At the bottom, there is a 'Sound Notification' toggle (Yes/No) and a 'Process 1. Input and Initial Results' button.

Figure 7. EPSG code input selection for georeferenced images.

7.2 Deep Learning Model Configuration and Options

A notable difference between the local and cloud models is that the local model includes post-processing of the predicted masks, which improved the quality of the detection. Note that the post-processing of the predicted masks in the cloud model is non-configurable and a limitation of the cloud model itself. As shown in Figure 8, users can choose between the two different deep learning models (local and cloud-based) for processing. Each model requires different inputs; selecting one automatically enables or disables the corresponding interface sections. Local processing is recommended, as it enables image analysis without limitations on image count or dependency on a stable internet connection.

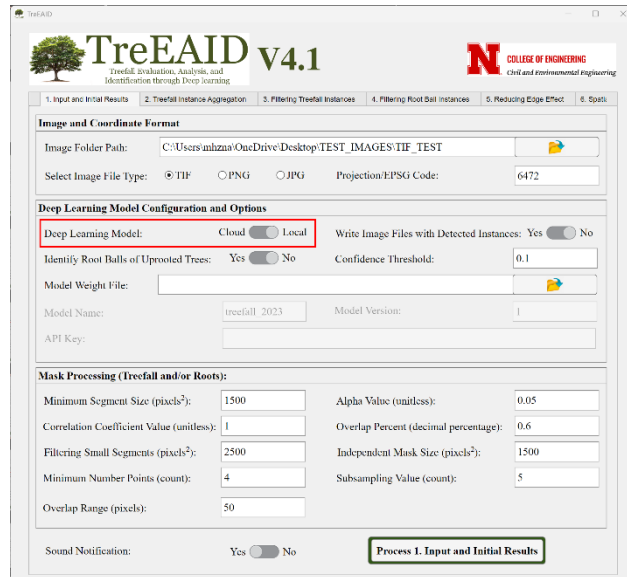


Figure 8. Interface section for selecting the type of deep learning model (cloud or local).

The next part of this tab is “Write Image Files with Detected Instances.” Data will be processed regardless of this option, but it is recommended to start by selecting 'Yes' to visualize a subset of results. This helps to verify whether the default parameters are appropriate and allow for necessary adjustments. Visualization can also help determine whether additional filtering is required; at the same time, users can assess the overall accuracy of the results. Once suitable threshold values have been identified, switching this option to “No” will reduce processing time. Figure 9 shows how to adjust this setting.

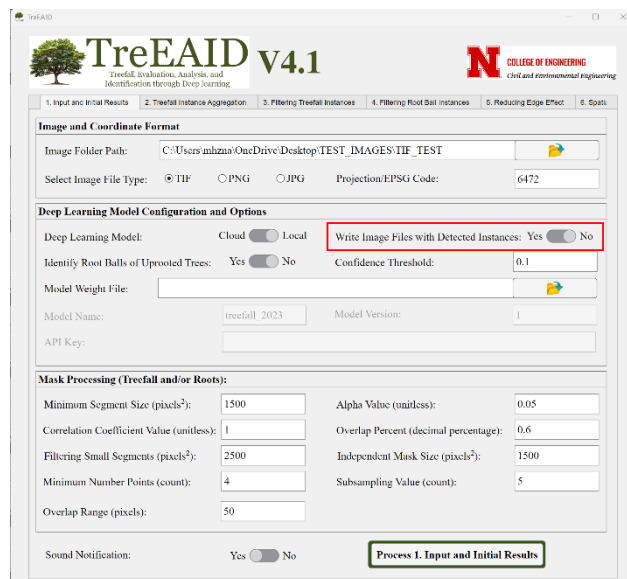


Figure 9. Option to specify whether image files with detected instances should be saved.

The next part of the tab is “Identify Root Balls of Uprooted Trees.” The software was initially designed to detect and identify the orientation of the fallen trees and generate a wind direction map as a representative of all predictions evaluated in the region. In addition to treefall direction analysis, the application can also evaluate the root ball of the uprooted trees upon the user's request, for a

comprehensive assessment. As shown in Figure 10, the user can specify their request for the root ball analysis in addition to the treefall evaluation. Selecting the “Yes” option activates root ball evaluation and records the results in a separate subfolder within the input folder. However, choosing the “No” option skips root ball assessments, which can reduce processing time.

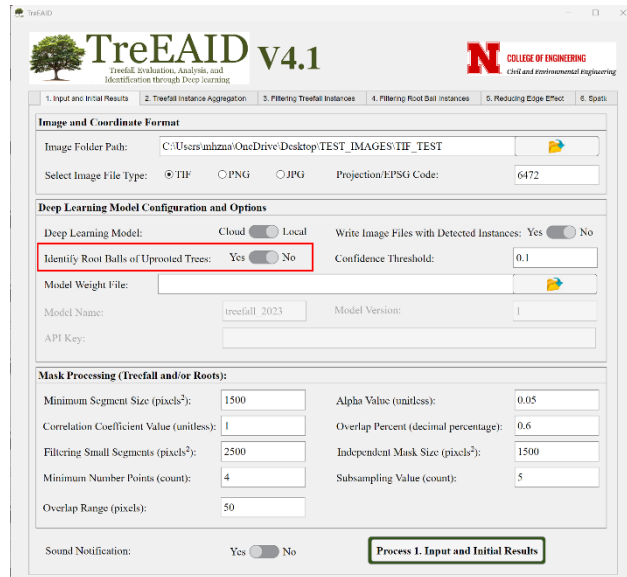


Figure 10. Option to include root ball evaluation with treefall vector analysis.

Opting for the local model means that all images will be processed on the user's local computer. To enable this, users must select the weight file included with the application. The file should have a .pt extension, as shown in Figure 11.

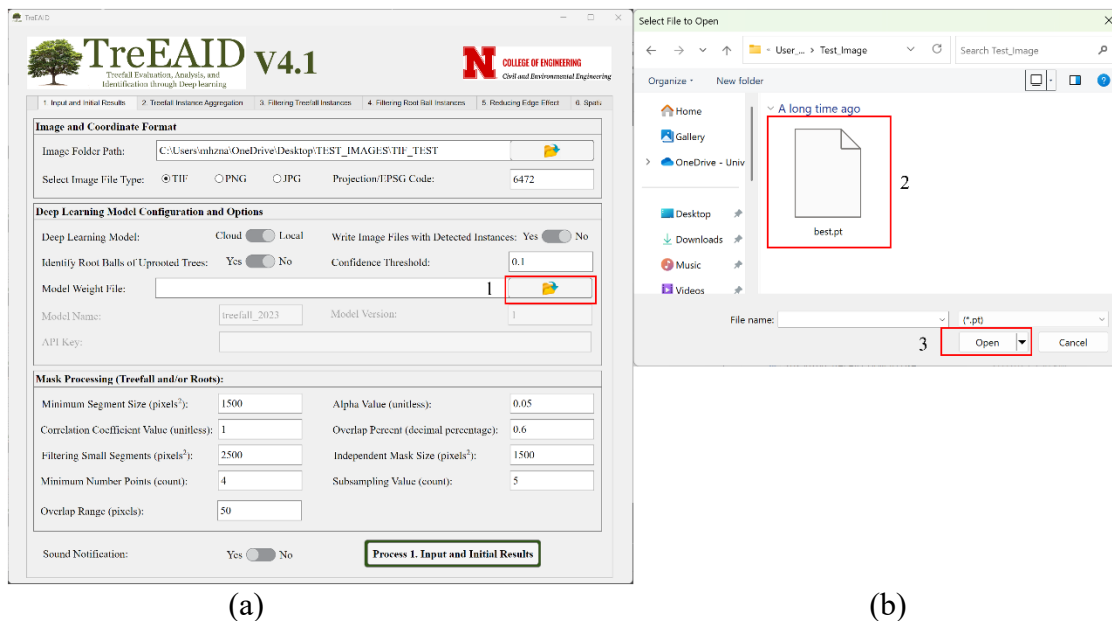


Figure 11. Weight file selection for local model processing: (a) locating the weight file; (b) selecting the shared weight file.

The confidence level indicates the probability that a detected object (either a fallen tree or a root ball) is correctly identified. This threshold value can be adjusted by users, as shown in Figure 12. Initially, setting this value to **10% (0.1) is recommended**. This threshold value applies to both treefall and root ball detections. By opting for a confidence level of 0.1, the model will consider all predictions identified as “treefall” and “root ball” with a probability higher than 10%. While this may seem low compared to other deep learning models, the YOLO models do not follow this trend. YOLO models can create valid predictions for detections with confidence levels as low as 5% [3]. Additionally, the goal is to capture all potential predictions in the first step; by setting a low threshold, the model includes detections even with low confidence, which can later be filtered using the 'Filtering Treefall Instances' tab.

The screenshot shows the TreEAID v4.1 application window. The title bar indicates it's a 'Treefall Evaluation, Analysis, and Identification through Deep Learning' tool. The interface has a tabbed menu at the top with six tabs: '1. Input and Initial Results', '2. Treefall Instance Aggregation', '3. Filtering Treefall Instances', '4. Filtering Root Ball Instances', '5. Reducing Edge Effect', and '6. Stats'. The first tab is active. Below the tabs, there are several sections for configuration:

- Image and Coordinate Format:** Includes fields for 'Image Folder Path' (C:\Users\mhznz\OneDrive\Desktop\TEST_IMAGES\TIF_TEST), 'Select Image File Type' (radio buttons for TIF, PNG, JPG), and 'Projection/EPSG Code' (6472).
- Deep Learning Model Configuration and Options:** This section contains:
 - 'Deep Learning Model' with 'Cloud' and 'Local' radio buttons.
 - 'Write Image Files with Detected Instances' with 'Yes' and 'No' radio buttons.
 - 'Identify Root Balls of Uprooted Trees' with 'Yes' and 'No' radio buttons.
 - 'Confidence Threshold' set to 0.1, which is highlighted with a red rectangular box.
 - 'Model Weight File' with a file path: C:\Users\mhznz\OneDrive\Desktop\TEST_IMAGES\TIF_TEST\best.pt.
 - 'Model Name' set to 'treefall_2023' and 'Model Version' set to '1'.
 - 'API Key' field.
- Mask Processing (Treefall and/or Roots):** A grid of input fields for various parameters:
 - Minimum Segment Size (pixels²): 1500
 - Correlation Coefficient Value (unitless): 1
 - Filtering Small Segments (pixels²): 2500
 - Minimum Number Points (count): 4
 - Overlap Range (pixels): 50
 - Alpha Value (unitless): 0.05
 - Overlap Percent (decimal percentage): 0.6
 - Independent Mask Size (pixels²): 1500
 - Subsampling Value (count): 5
- At the bottom, there is a 'Sound Notification' section with 'Yes' and 'No' radio buttons, and a 'Process 1. Input and Initial Results' button.

Figure 12. Confidence threshold value definition for treefall and root ball detections.

For the cloud-based model, the weight file is hosted online, eliminating the need for users to select the weight file for processing. Nevertheless, providing the most up-to-date model information remains crucial, as depicted in Figure 13. Prior to using the application with the cloud-based functionality, it is important to contact Prof. Richard Wood at rwood@unl.edu to obtain the API key and most recent model version.

TreEAID v4.1
Treefall Evaluation, Analysis, and Identification through Deep Learning

1. Input and Initial Results 2. Treefall Instance Aggregation 3. Filtering Treefall Instances 4. Filtering Root Ball Instances 5. Reducing Edge Effect 6. Spots

Image and Coordinate Format

Image Folder Path: C:\Users\mlh\OneDrive\Desktop\TEST_IMAGES\TIF_TEST

Select Image File Type: ☒ TIF ☐ PNG ☐ JPG Projection/EPSSG Code: 6472

Deep Learning Model Configuration and Options

Deep Learning Model: Cloud ☒ Local ☐ Write Image Files with Detected Instances: Yes ☐ No ☒

Identify Root Balls of Uprooted Trees: Yes ☐ No ☒ Confidence Threshold: 0.1

Model Weight File: C:\Users\mlh\OneDrive\Desktop\TEST_IMAGES\TIF_TEST\best.pt

Model Name: treefall_2023 Model Version: 1

API Key: [REDACTED]

Mask Processing (Treefall and/or Roots):

Minimum Segment Size (pixels²): 1500 Alpha Value (unitless): 0.05

Correlation Coefficient Value (unitless): 1 Overlap Percent (decimal percentage): 0.6

Filtering Small Segments (pixels²): 2500 Independent Mask Size (pixels²): 1500

Minimum Number Points (count): 4 Subsampling Value (count): 5

Overlap Range (pixels): 50

Sound Notification: Yes ☐ No ☒ **Process 1. Input and Initial Results**

Figure 13. Providing model information for cloud-based evaluation.

The 'Mask Processing' section enhances detection quality of the local segmentation model by addressing two common challenges in instance segmentation [4]: (1) fragmented masks and (2) duplicated detections. These issues are particularly relevant for objects with complex geometry, such as fallen trees and root balls in post-disturbance environments.

1) Multi-segment masks (fragmented output for a single object):

Instance segmentation models may predict a single object (e.g., a tree or root ball) as multiple disconnected mask fragments. As shown in Figure 14b, a single tree may be represented by two spatially separated regions. In YOLO-based segmentation, multiple disconnected fragments can be produced, but due to export limitations, especially in formats that retain only a single polygon (typically the largest or first), secondary fragments may be lost. This occurs because YOLO does not support holes or multiple polygons per object. The loss of such fragments leads to underestimation of object dimensions, such as total tree length or root ball width. These metrics are crucial for feature-based analyses like, taper rate estimation. To address this, the application's local post-processing pipeline merges fragmented masks when they are collinear, and treats them as separate detections when they are not. This refinement is governed by several user-defined parameters, listed in Table 5:

- Minimum Segment Size
- Alpha Value
- Correlation Coefficient
- Overlap Percent
- Filtering Small Segments
- Independent Mask Size
- Minimum Number of Points
- Subsampling Value
- Mask Overlap Range

2) Multi-detection of a single instance (duplicated predictions):

In some cases, the same object may be detected multiple times. For example, two overlapping detections may be generated for the same fallen tree, as illustrated in Figure 14a. This leads to inflated counts and misinforms analyses that rely on accurate object quantification, especially in some treefall-based damage assessments [5]. To reduce redundancy, the application applies a Non-Maximum Suppression (NMS), similar to a filter at the mask level, using user-defined thresholds for Intersection over Union (IoU) and confidence level. The percent of area overlap (IoU) determines whether two masks are duplicates. If this threshold is exceeded, only one mask is retained, typically the one with higher confidence or greater area.

Although the YOLO framework incorporates basic post-processing mechanisms—such as filtering small masks and applying NMS based on bounding boxes [6], these are often insufficient for treefall and root ball applications. For example:

- Collinear fragments of a single tree may be misclassified as distinct objects.
- Closely spaced trees lying parallel may be incorrectly merged.
- Irregular shapes, like root balls, are poorly captured by bounding box-level NMS [7]

Moreover, YOLO's NMS is applied to rectangular bounding boxes, not the actual mask shapes, limiting its precision in environments with overlapping, elongated, or curved objects (e.g., tornado-impacted forests) [8].

To overcome these limitations, the Mask Processing (Treefall and/or Roots) module in the Input and Initial Results tab (Tab 1) employs a custom, geometry-driven post-processing pipeline (Figure 15).:Figure 15. This approach enhances segmentation masks by:

- Merging collinear segments using Alpha Value (angle threshold) and Correlation Coefficient (linear fit).
- Avoiding false merging of neighboring trees using Mask Overlap Range.
- Filtering noise via Minimum Segment Size and Filtering Small Segments.
- Separating non-collinear fragments using Independent Mask Size.
- Prioritizing best detections based on overlap (Overlap Percent), area, and confidence level.
- Reducing processing time using Subsampling Value for treefall mask points.
- Ensuring geometric completeness by retaining only predictions with a minimum Number of Points on the mask boundary.

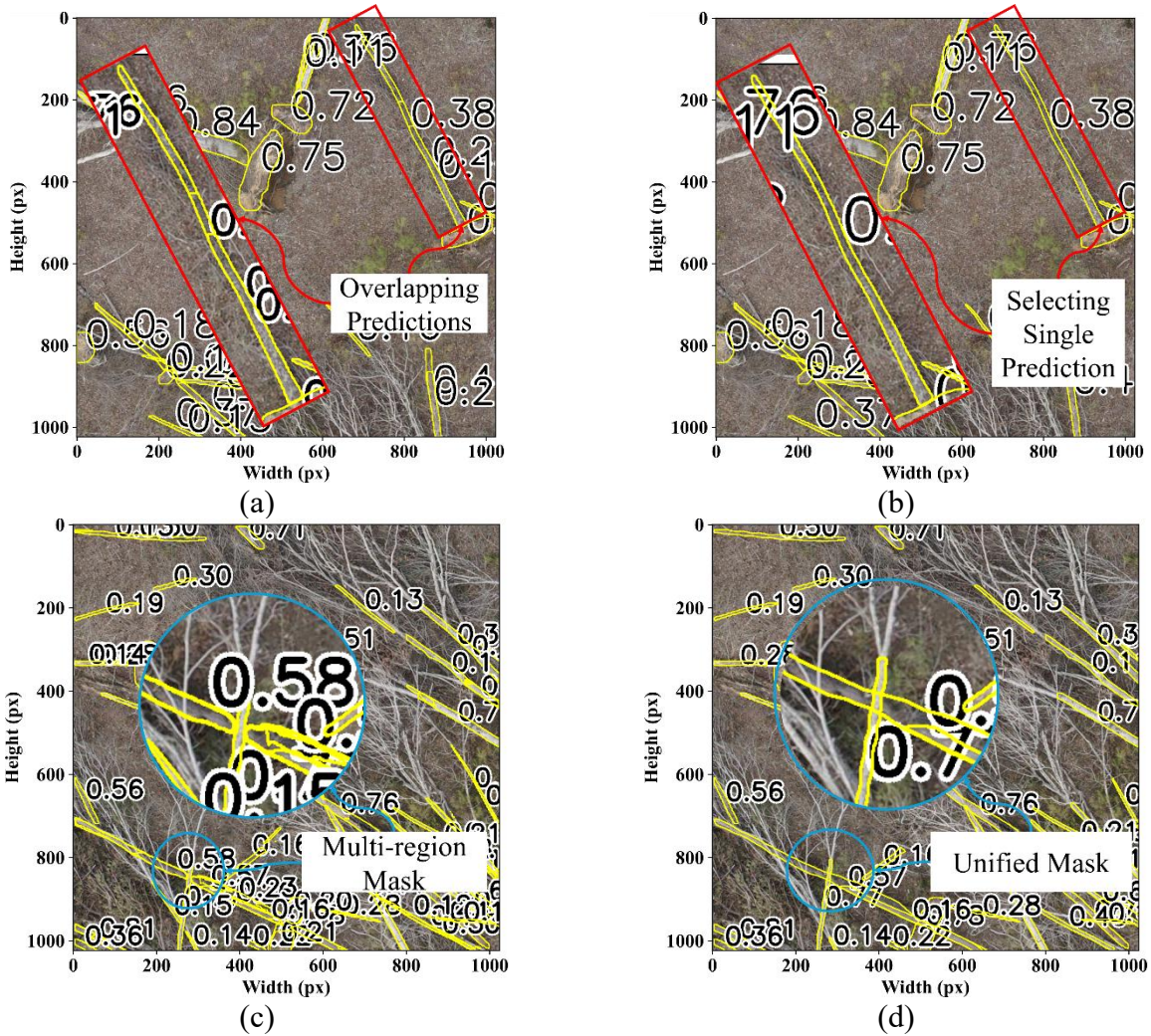


Figure 14. Mask refinement results: (a) overlapping predictions before post-processing; (b) duplicated prediction resolved by retaining the largest mask; (c) fragmented mask segments for a single object; (d) post-processing result with unified mask.

As shown in Figure 15, this section defines nine parameters used to refine instance segmentation masks for treefall and root ball detection. Using these parameters, the application performs several operations on the mask segments within fragmented masks and in cases of multiple detections of a single instance (i.e., duplicated predictions). It removes small segments considered noise, merges collinear segments belonging to the same tree, and separates non-collinear segments. It is important to note that the default values defined for the parameters are based on the data analyzed by the developers (with a GSD of approximately 2 cm), so users may need to optimize these values based on their dataset's characteristics, including GSD, image quality, and tree properties (species, age, thickness, length, etc.). The starting values for mask refinement are specified in Table 5. These parameters are defined as follows:

Table 5. Starting values for mask refinement parameters.

Parameters	Unit	Values
Minimum segment size	pixels ²	< 1500
Alpha value	unitless	0.05
Correlation coefficient	unitless	> 0.80
Overlap percent	decimal percentage	> 0.60
Filtering small segments	pixels ²	< 2500
Independent mask size	pixels ²	> 1500
Minimum number of Points	count	≥ 4
Subsampling value	count	5
Mask Overlap Range	pixels	> 50

- Minimum Segment Size [pixels²]: This parameter removes small mask sections considered noise during the initial detection. It helps to improve the accuracy of treefall or root ball identification by filtering out insignificant fragments.
- Alpha Value [unitless]: Alpha value defines the connection between large sections of fragmented masks. This value helps to connect multi-region root balls and multi-region and linear treefall sections when the components are part of the same object. It is used to generate a unified mask from multiple segments of the same object.
- Correlation Coefficient Value [unitless]: This parameter measures the Pearson Correlation Coefficient value of “r” between fragmented mask segments to assess their linear alignment, used only for treefall instances. A higher correlation indicates that the segments likely belong to the same fallen tree and may be merged into a single instance.
- Overlap Percent [percent]: This parameter assesses the overlap between the predicted masks in both treefall and root ball. When two masks significantly overlap, they are considered duplicated detections of the same object, and only the one with the largest area is retained for further processing.
- Filtering Small Segments [pixels²]: In multi-region masks with no collinearity, this parameter eliminates disconnected mask fragments too small to be considered in the analysis.
- Independent Mask Size [pixels²]: This parameter is used to determine whether a segment within a fragmented mask should be retained as a separate instance. If segments are non-collinear, only sufficiently large fragments are retained.
- Minimum Number of Points [count]: This parameter specifies the minimum number of coordinate points required to define a mask to be retained in the final outputs. The appropriate value depends on the geometric representation of the detected instance. For example, modeling a root ball as an ellipse requires at least five points to estimate the five parameters

of an ellipse equation. Similarly, representing a fallen tree trunk as a rectangular polygon requires at least four coordinate points.

- **Subsampling Value [count]:** This parameter aims to reduce the coordinate points along each detected fallen tree mask. Reducing the number of points in each prediction reduces file size and computational load, making predictions easier to store and process.
- **Mask Overlap Range [pixels]:** This parameter defines the minimum pixel overlap required to consider two mask segments as potentially representing the same object. If the projection between two segments in either X or Y coordinates exceeds this threshold, the segments are not evaluated for merging. The primary goal ensures that parallel mask sections are not mistakenly merged as a single tree.

The screenshot shows the TreEAID v4.1 software interface. The top navigation bar includes tabs for: 1. Input and Initial Results, 2. Treefall Instance Aggregation, 3. Filtering Treefall Instances, 4. Filtering Root Ball Instances, 5. Reducing Edge Effect, and 6. Spots. The main window is divided into three main sections. The first section, 'Image and Coordinate Format', includes fields for 'Image Folder Path' (C:\Users\mhznz\OneDrive\Desktop\TEST_IMAGES\TIF_TEST), 'Select Image File Type' (TIF, PNG, JPG), and 'Projection/EPSPG Code' (6472). The second section, 'Deep Learning Model Configuration and Options', includes fields for 'Deep Learning Model' (Cloud, Local), 'Write Image Files with Detected Instances' (Yes, No), 'Identify Root Balls of Uprooted Trees' (Yes, No), 'Confidence Threshold' (0.1), 'Model Weight File' (C:\Users\mhznz\OneDrive\Desktop\TEST_IMAGES\TIF_TEST\best.pt), 'Model Name' (treefall_2023), 'Model Version' (1), and 'API Key'. The third section, 'Mask Processing (Treefall and/or Roots)', is highlighted with a red box and contains the following parameters: 'Minimum Segment Size (pixels²)' (1500), 'Alpha Value (unitless)' (0.05), 'Correlation Coefficient Value (unitless)' (1), 'Overlap Percent (decimal percentage)' (0.6), 'Filtering Small Segments (pixels²)' (2500), 'Independent Mask Size (pixels²)' (1500), 'Minimum Number Points (count)' (4), 'Subsampling Value (count)' (5), and 'Overlap Range (pixels)' (50). At the bottom, there is a 'Sound Notification' section with 'Yes' and 'No' options, and a 'Process 1. Input and Initial Results' button.

Figure 15. Filtering parameter settings for refining instance segmentation masks

The application, by default, is designed to notify the user about the status of processing through voice notification designated in each processing Tab. However, this feature can be disabled by selecting “No” option in each tab as shown in Figure 16.

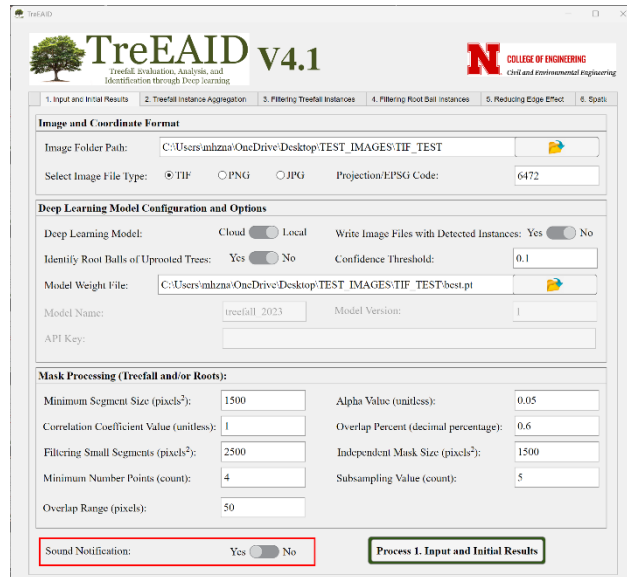


Figure 16. Option to enable or disable sound notification after processing.

After selecting this option, users can begin processing images to obtain the results. Processing time depends on the number of images and the number of treefall and root ball instances per image. For example, processing 3,842 images from a survey region covering an area of 1.07 km² at an approximate GSD of 2.0 cm resulted in the identification and processing of 1,723 root ball instances and 9,424 treefall instances across the dataset. The total processing time was approximately 6 hours and 35 minutes. Upon completion, users receive a notification via a pop-up and, if enabled, a configurable sound alert. If an error occurs, processing stops and the user is notified. Figure 17 shows the process completion notification. For errors, refer to the 'Error Messages' section for potential causes and suggested solutions. If the error is not listed in the appendix, users are encouraged to contact Prof. Richard L. Wood (rwood@unl.edu) or Mitra Nasimi (mnasimi2@huskers.unl.edu), including the error message and relevant data.

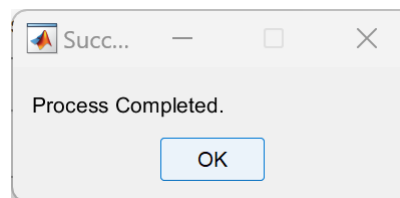


Figure 17. Notification displayed upon successful completion of the processing step

8 Treefall Instance Aggregation

The “Treefall Instance Aggregation” section aims to reduce duplicated predictions for the same fallen tree. This issue often arises when obstacles or visual discontinuities in the imagery cause a single tree to be detected as multiple, fragmented segments. Although the post-processing of initial detections improves the size and number of detected object masks, it primarily targets multi-region (fragmented) masks and duplicated masks with significant overlap.

This tab aims to identify the instances of the fallen tree polygons that are adjacent to each other but have no or minimal overlap. If they meet specific criteria, these fragmented predictions can be merged to reconstruct an extended and aggregated portion of the tree trunk, as demonstrated later in this section. Root ball instances are less likely to have multiple predictions at this stage; therefore, aggregation processing is applied only to fallen trees. Figure 18 shows the output where multiple predictions for a tree instance are converted into a single prediction or aggregation. Note that this tab is optional, and it processes data to achieve an accurate number of the tree instance vectors. Performing this step is beneficial since some of the treefall methods are sensitive to the number of fallen trees [5].

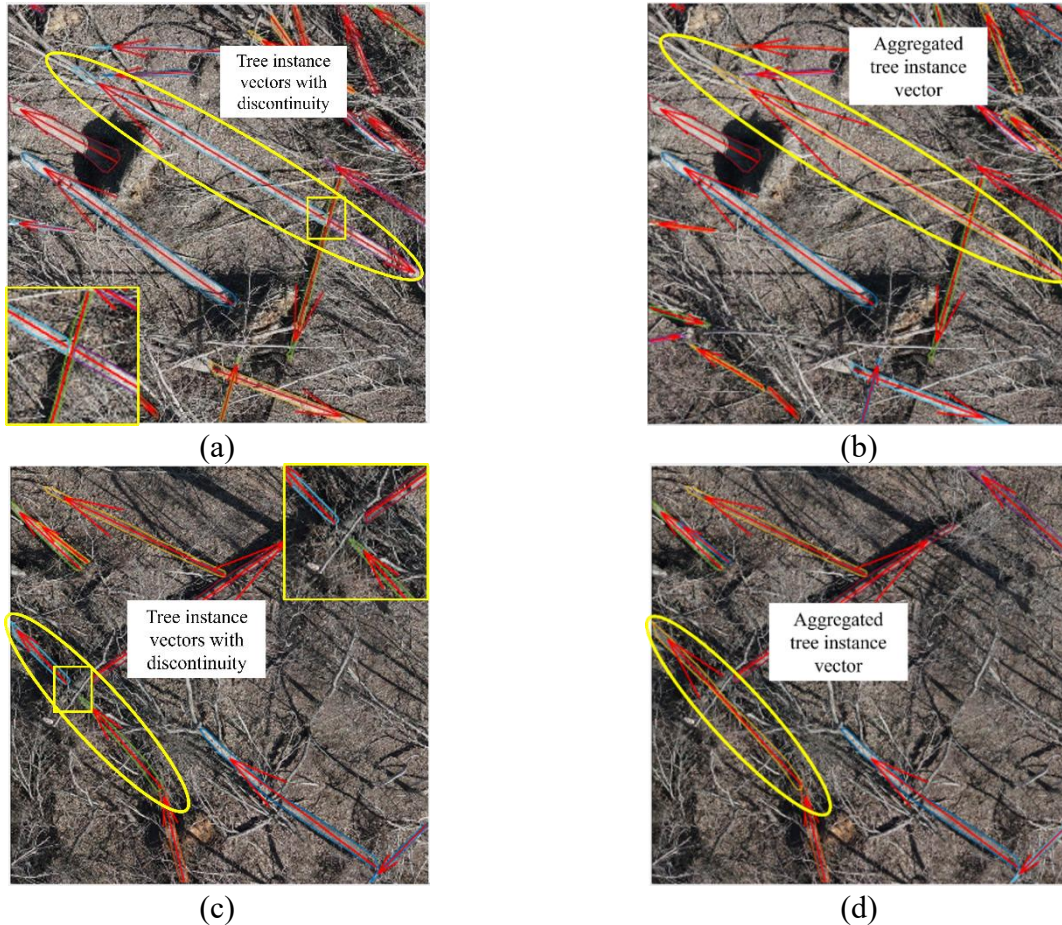


Figure 18. Treefall aggregation example: (a) vectors disconnected by a fallen trunk; (b) aggregated polygon and vector; (c) vectors separated by a root ball; (d) aggregated polygon and vector.

8.1 Aggregation Parameter

Users must identify threshold values for the parameters identified in this section to start the processing. Similar to previous steps, it is recommended to process a subset of the input data by choosing “Yes” to visualize the results for that subset. This helps to verify whether the default parameters are appropriate and allows for necessary adjustments. Once suitable threshold values are identified, switching this option to “No” will reduce processing time during full-batch processing. Users can visualize the results of the processing by selecting the “Yes” Option for the “Write Image Files with Detected Instance,” as shown in Figure 19.

The screenshot shows the TreEAID v4.1 software window. The title bar says 'TreEAID'. The main header includes a tree logo, 'TreEAID v4.1', and 'Treefall Evaluation, Analysis, and Identification through Deep learning'. On the right, there's a logo for 'N COLLEGE OF ENGINEERING Civil and Environmental Engineering'. Below the header is a tabbed interface with six tabs: '1. Input and Initial Results', '2. Treefall Instance Aggregation', '3. Filtering Treefall Instances', '4. Filtering Root Ball Instances', '5. Reducing Edge Effect', and '6. Spatial'. The '2. Treefall Instance Aggregation' tab is active. Inside this tab, there's a section titled 'Aggregation Parameters (Treefall):'. It contains several parameters with input fields or toggle switches: 'Write Image Files with Detected Instances:' (Yes/No toggle, with 'Yes' selected and highlighted by a red box), 'Reprocess Based on Tree Taper:' (Yes/No toggle, with 'Yes' selected), 'Length of Extension [L]*:' (input field with '3'), 'Tree Thickness Multiplier (unitless):' (input field with '1'), 'Tree Overlap Range [L]*:' (input field with '2'), 'Angle Difference (deg):' (input field with '10'), and 'Percent of Interior Extension (decimal percentage):' (input field with '0.5'). At the bottom of this section, there's a note: '* Units align with the global coordinate system defined for TIFF and PNG (e.g., meters). Otherwise, units are in pixels.' Below the parameters section is a 'Sound Notification:' section with a 'Yes/No' toggle (set to 'No') and a green button labeled 'Process 2. Tree Instance Aggregation and Results'.

Figure 19. Option to specify whether image files with detected instances should be saved on the mask refinement and results tab.

The outcome of this step is, by default, recorded in a table of geotagged (aggregated) treefall vectors and updated prediction masks. The direction of the aggregated treefall vector is computed as the weighted average of the selected instances. The process of computing the weighted average follows these steps:

- 1) The direction of each fallen tree is determined using its horizontal orientation angle. During this step, all directional values are constrained to the range of 0° to 180° , representing a general east–west orientation regardless of the precise azimuth. This simplification ensures consistency in directional assessments.
- 2) Within each group of treefall instances, defined as multiple predictions or fragments that lie within each other’s region of interest and meet the aggregation criteria, the first instance’s horizontal direction is selected as the reference. This reference is assigned a directional sign of +1, corresponding to its orientation angle, denoted as (θ_{href}) .

- 3) Each remaining instance in the group is assigned a directional sign of either +1 or -1, based on its angular deviation from the reference orientation, as defined in Equation 1. This sign assignment eliminates directional magnitude bias and ensures consistency when aggregating vectors that may represent the same tree.

$$Sign = \begin{cases} +1, & \text{if } \theta_i = |\theta_i - \theta_{href}| \leq 45 \\ -1, & \text{if } \theta_i = |\theta_i - \theta_{href}| > 45 \end{cases} \quad \text{Equation 1}$$

- 4) The overall Horizontal Angle Sign (HAS) is calculated using a weighted average that incorporates the directional sign (from Step 3), the taper rate, the segment length, and the confidence level. This formulation ensures that stronger, more reliable predictions have greater influence. Equation 2 defines this relationship.

$$HAS = \frac{\sum \text{Confidence level} \times \text{length} \times \text{taper rate} \times \text{sign}}{\sum \text{Confidence level} \times \text{length} \times \text{taper rate}} \quad \text{Equation 2}$$

- 5) Finally, the overall orientation angle θ_{final} for the aggregated treefall vector is determined based on the sign of the HAS value. Equation 3 applies the sign to determine whether to use the mean orientation of positively signed or negatively signed predictions:

$$\theta_{final} = \begin{cases} \text{mean}(\theta_i | \text{sign}_i = 1), & \text{if } HAS \geq 0 \\ \text{mean}(\theta_i | \text{sign}_i = -1), & \text{if } HAS < 0 \end{cases} \quad \text{Equation 3}$$

In this step, outputs can also be optionally generated based on the taper rate, leveraging the updated treefall masks (i.e., treefall polygons) created during aggregation. Users may enable this option by selecting the “Yes” option for “Reprocess Based on Tree Taper”, as shown in Figure 20. It is important to note that enabling this option will increase the processing time, as the system recalculates taper rates and fall directions based on the aggregated mask data [1]. Despite the increase in processing time, it is generally recommended to enable this, given the observed benefit of the treefall aggregation step.

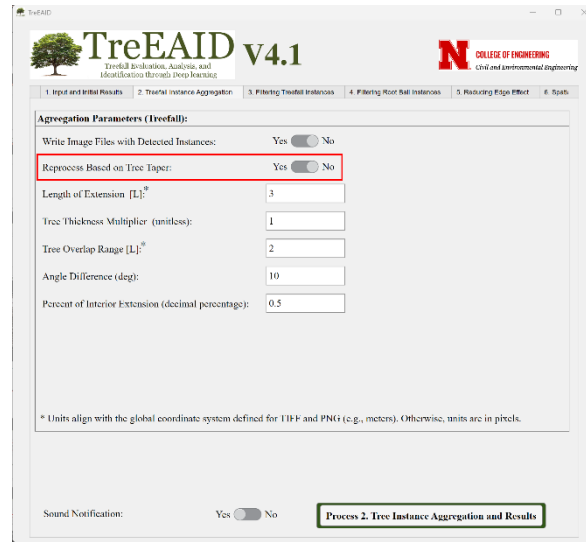


Figure 20. Option to reprocess results based on tree taper using aggregated mask information.

As shown in Figure 21, five user-defined parameters control the initiation of the treefall instance aggregation process. Based on these threshold values, the application evaluates treefall vectors within each image and determines whether adjacent vectors represent fragmented portions of the same fallen tree. If the criteria are satisfied, the vectors are merged into a single instance. This aggregation step is especially important when results must represent each unique treefall occurrence as a single vector, as required by certain downstream treefall analysis methods [5].

It is important to note that the default values defined for the parameters are based on the data analyzed by the developers with a ground sample distance (GSD) of approximately 2 cm. Users working with imagery of different GSDs, qualities, or vegetation characteristics (e.g., species, age, thickness, length) may need to adjust these values accordingly. The starting defaults for these aggregation parameters are provided in Table 6 , with each parameter defined as follows:

Table 6. Starting threshold values for aggregation parameters.

Parameters	Unit	Values
Length of Extension	Global coordinate system (e.g., meters) or pixels (JPG)	3
Tree Thickness Multiplier	unitless	1
Tree Overlap range	Global coordinate system (e.g., meters) or pixels (JPG)	2
Angle difference	degree	10
Percent of Interior Extension	Decimal percentage	0.5

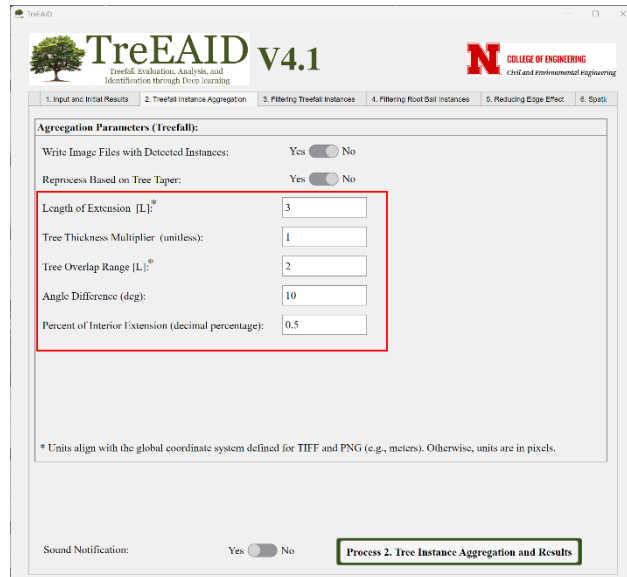


Figure 21. Treefall instance aggregation parameters used to generate regions of interest and assessment.

The parameters in this tab are designed to identify segments of the same fallen tree within a single image. To accomplish this, a region of interest (ROI) is generated at both ends of each detected tree polygon. The corner coordinates of each ROI polygon are calculated based on the user-defined parameter values described below.

- **Length of Extension ($L_{\text{extension}}$) [m]:** This parameter defines the outward extension of the region of interest (ROI) from each end of the fallen tree, as aligned with the tree. It accounts for gaps or discontinuities in the tree trunk that may result from occlusion, fragmentation, or reduced visibility in the image. The ROI enables the algorithm to assess whether disconnected segments may belong to the same tree. The length of this extension is user-defined and is applied symmetrically at both ends of the tree polygon.
- **Tree Thickness Multiplier (N) [unitless]:** This parameter is a user-defined scaling factor that determines the lateral width of the ROI used in the analysis. It scales the ROI width based on the predicted tree's thickness (t), defined as the median distance between the sides of the tree polygon. The ROI is centered on the treefall vector and extends symmetrically and perpendicularly as a function of N and t . For example, if $N = 2$, the ROI extends $2 \times t$ on each side, for a total lateral ROI width of $4 \times t$. This scaling keeps the ROI proportional to tree geometry, preventing it from being too narrow (missing nearby segments) or too wide (introducing irrelevant data, especially for parallel treefall matches).
- **Tree Overlap Range [m]:** This parameter is used to prevent the erroneous merging of parallel but distinct trees. For two segments that may belong to the same tree but are separated by a gap, their projections along the X and Y axes are evaluated. If the overlap between two tree prediction vectors exceeds a user-defined threshold in either direction, the segments are considered separate trees—regardless of visual proximity—and are excluded from merging. This ensures that parallel, closely aligned trees are not mistakenly treated as parts of a single instance. This threshold is defined in the same units as tree length).

- Angle Difference [deg]: This threshold defines the maximum allowable angular deviation between two treefall segments being evaluated for aggregation. If the angle between the orientation vectors of two segments exceeds this threshold, they are treated as separate instances. This ensures that only tree segments with consistent fall directions are considered parts of the same fallen tree.
- Percent of Interior Extension (PIE) [decimal percentage]: This parameter defines the extent to which the region of interest (ROI) extends inward from each end of the detected fallen tree trunk. PIE is specified as a decimal value representing a percentage of $L_{\text{extension}}$. For example, 0.5 corresponds to 50% of $L_{\text{extension}}$. This inward extension enables the detection and aggregation of treefall segments with slight overlaps that might otherwise be missed.

$$\text{Interior Extension} = L_{\text{extension}} \times \text{PIE}$$

Equation 4

In this equation, $L_{\text{extension}}$ is the user-defined outward extension length, and PIE is the Percent of Interior Extension, also defined by the user.

Figure 22 illustrates all the ROI generating parameters defined in this section in detail. The red polygons represent the exterior ROI, while the blue polygon indicates the interior ROI extension. The dashed green line represents the tree polygon centerline (prediction vector).

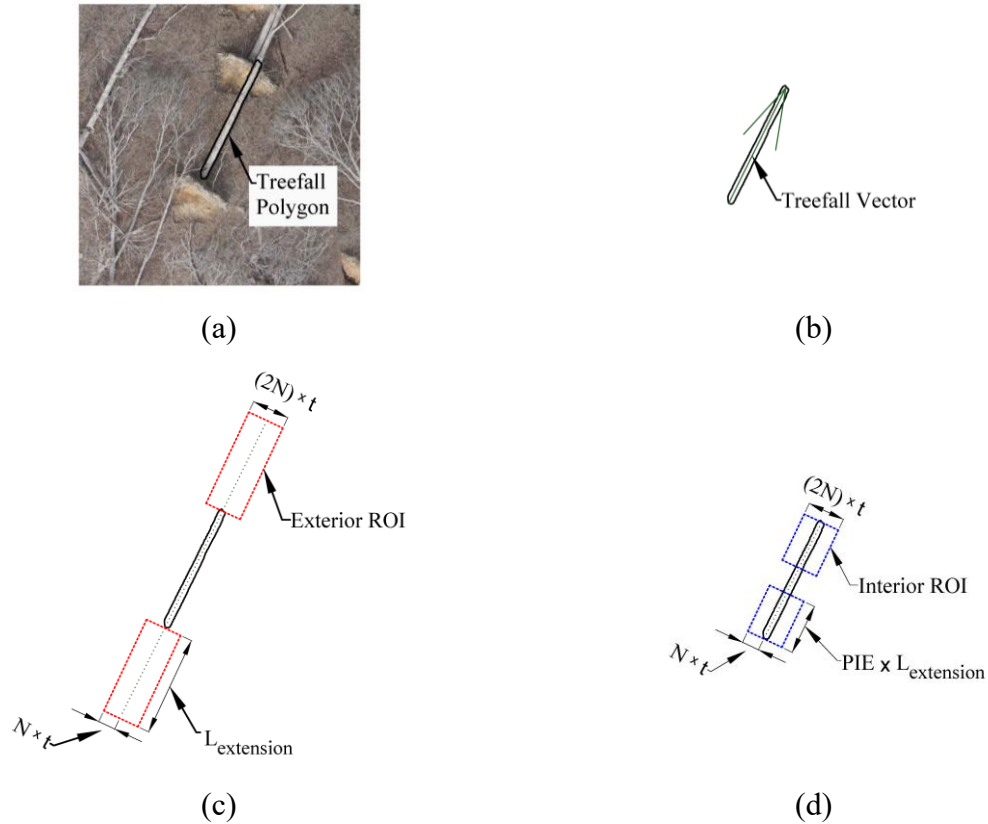


Figure 22. Region of interest (ROI) parameters in tree instance aggregation: (a) treefall polygon on the orthophoto; (b) estimated treefall vector; (c) exterior ROI with length of extension ($L_{\text{extension}}$) and width scaling $N \times t$; and (d) interior ROI defined by $\text{PIE} \times L_{\text{extension}}$ and width scaling.

9 Filtering Treefall Instances

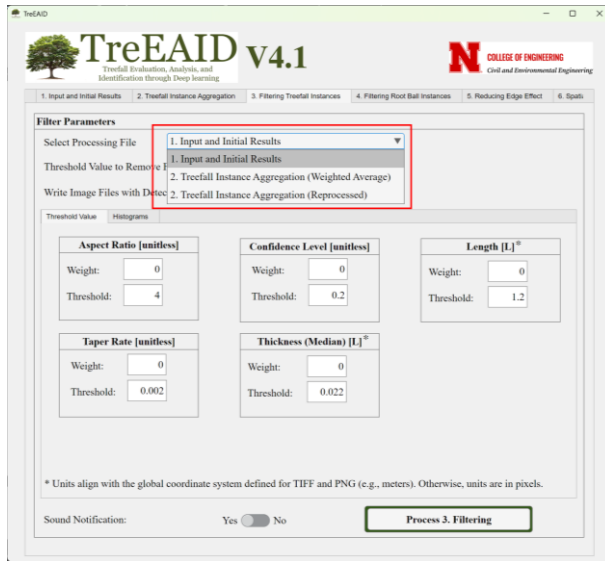
The Filtering Treefall Instances section is designed to remove inaccurate predictions using feature-based parameters. These parameters help eliminate a substantial number of false positives while preserving accurate predictions. This step enhances the interpretability of the deep learning model by applying physically meaningful filters derived from tree characteristics.

9.1 Filter Parameters

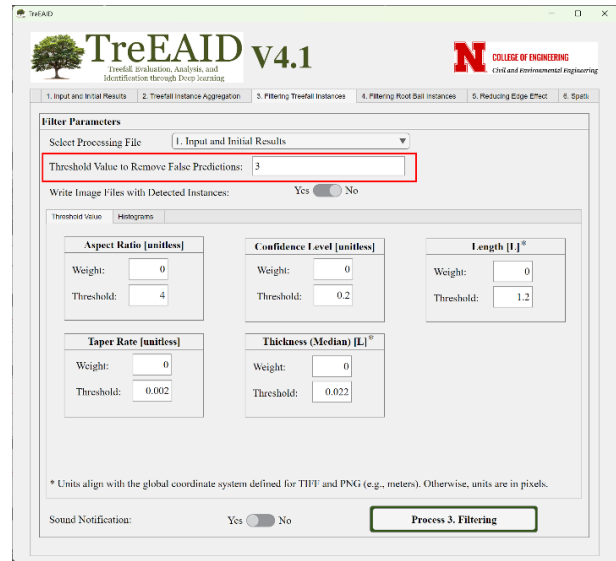
The application uses a file-mediated pipeline. Therefore, users must select the output from a prior processing step to serve as the input dataset for this stage. Alongside the selected dataset, users must define threshold and weight values to filter out false predictions. The three processing modes available in this section are as follows:

- “1. Input and Initial Results”: Filters the outputs generated from the *Input and Treefall Results* tab.
- “2. Tree Instance Aggregation (Weighted Average)”: Filters the outputs produced by the *Treefall Instance Aggregation* tab using a weighted averaging approach. The aggregation combines initial treefall polygons based on spatial proximity and specific criteria. The direction of each aggregated tree instance is computed from the weighted average of individual predictions, with weights determined by confidence level, taper rate, and length.
- “2. Tree Instance Aggregation and Results”: Filters outputs generated by the *Treefall Instance Aggregation* tab (Reprocessed version). In this case, the direction of the aggregated treefall vector is estimated based on the taper rate of the final aggregated polygon, rather than a weighted average.

As shown in Figure 23, predictions are filtered according to a combination of feature-based parameters and a user-defined threshold, "Threshold Value to Remove False Predictions." Each parameter can be assigned a weight, and any prediction whose combined weighted value exceeds the threshold will be considered inaccurate and removed. The threshold value can be adjusted to control the strictness of filtering. To further refine the filtering process, users may assign non-uniform weights to individual parameters based on their perceived reliability or significance. Similar to previous steps, visualization options are available to review results interactively and validate the effectiveness of chosen filter settings.



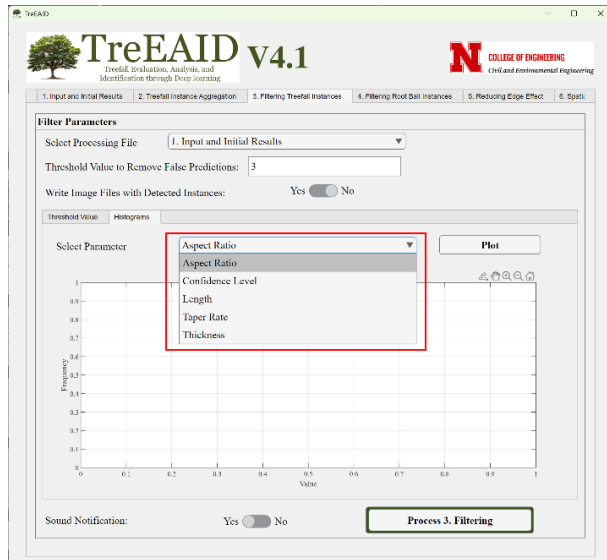
(a)



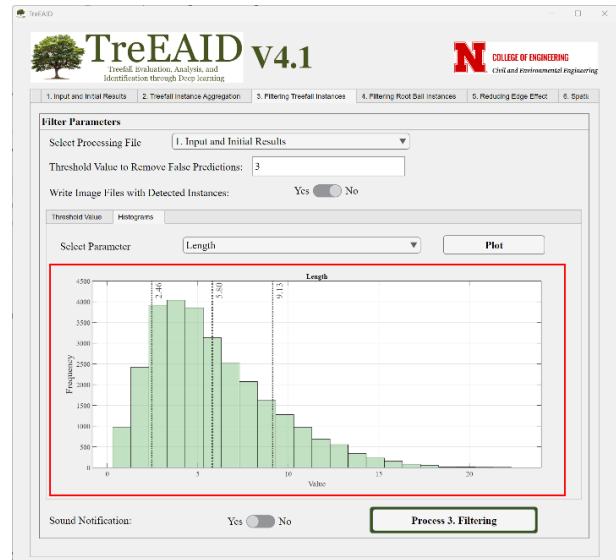
(b)

Figure 23. Filtering treefall instances: (a) selecting the dataset to filter; (b) setting the threshold value to eliminate false predictions.

The filter parameters defined in this section include aspect ratio, confidence level, length, taper rate, and thickness. These parameters are measured in the global coordinate system (e.g., meters) for PNG and TIF images, and at the pixel scale for JPG images. Parameter ranges can be adjusted based on the image resolution and the distribution of values, as visualized in the histogram section of the tab as shown in Figure 24. Weight values may also be modified to match the desired sensitivity of the filtering process. Definitions for each parameter are provided in the following section.



(a)



(b)

Figure 24. Visualization of parameter distributions: (a) selection of filtering parameter; (b) histogram showing distribution of aspect ratio with mean and standard deviation.

Various geometric parameters related to the physical characteristics of the predicted treefalls were assessed during the parametric analysis. These filters were designed to eliminate physically

implausible or abnormally shaped detections. The analysis showed that a combination of parameters effectively removed inaccurate predictions while retaining valid instances. It is important to note that these filtering parameters were calibrated using a dataset with a ground sampling distance (GSD) of approximately 2 cm. Users may need to determine optimized values for their own datasets, depending on factors such as GSD, image quality, and tree characteristics (e.g., species, age, thickness, and length). Table 7 summarizes the initial threshold values defined for each filtering parameter.

- **Aspect Ratio [unitless]**: The aspect ratio quantifies the elongation of a treefall prediction by comparing its length (L) to its thickness (t). This is a unitless parameter calculated as:

$$AR = \frac{L}{t} \quad \text{Equation 5}$$

Where L is the length of the predicted tree polygon (in meters or pixels, depending on the image format) and t is the median thickness of the tree polygon. For example, when the aspect ratio (AR) equals 4, a shape-based filtering is applied: polygons with **AR < 4** are classified as too short or stubby and excluded, while those with **AR > 4** are retained, as they likely represent taller treefall instances. [9].

- **Confidence Level**: This parameter represents the model-assigned probability that a predicted treefall instance is correct. Expressed as a decimal percentage (e.g., 20% is written as 0.20), it reflects the classification certainty produced by the segmentation or detection model. Higher values indicate greater confidence in the prediction.
- **Length [m]**: The length parameter represents the longitudinal extent of the predicted tree trunk polygon. Here the units depend on the image type: meters (or feet) for TIF and PNG images (aligned with global coordinates), and pixels for JPG images.
- **Taper Rate [unitless]**: Taper rate refers to the mean slope of the trunk along its primary axis, calculated between the start and end points of the polygon. This unitless parameter can help detect inconsistencies in tree shape that result in low tree taper.

$$TR = \frac{t_{base} - t_{end}}{L} \quad \text{Equation 6}$$

- **Thickness (median)**: This parameter represents the median thickness (i.e., width) of the treefall polygon across its span. Like length, the unit is in meters (or feet) for TIF and PNG formats and pixels for JPG images. It provides a robust estimate of tree width, useful for validating the physical plausibility of the predicted treefall instance.

Table 7. Starting threshold values for defining filter parameters.

Parameters	Symbol	Values	Units
Aspect Ratio	AR	< 4	$\frac{m}{m}$ [unitless]
Confidence Level	CL	< 0.2	Decimal percentage [unitless]
Length	L	< 1.2	Global coordinate system (e.g., meters) or pixels (JPG)
Taper Rate	TR	< 0.002	$\frac{m}{m}$ [unitless]
Thickness (median)	t	< 0.022	Global coordinate system (e.g., meters) or pixels (JPG)

10 Filtering Root Ball Instances

The Filtering Root Ball Instances section aims to remove erroneous predictions using several feature-based parameters. This filtering process is designed to enhance the interpretability of the deep learning model by relying on the physical characteristics of predicted root balls.

10.1 Filter Parameters

Users must select the output from the Input and Initial Results tab as the input dataset for this stage, along with a user-defined threshold value to eliminate false predictions. As shown in Figure 25, root ball instances are filtered based on whether the combined weighted score of specified parameters exceeds the defined threshold. Each parameter can be assigned a weight based on its perceived reliability. Predictions are excluded if their combined weighted score exceeds the user-defined threshold. As in previous steps, users can also visualize results during processing to assess performance.

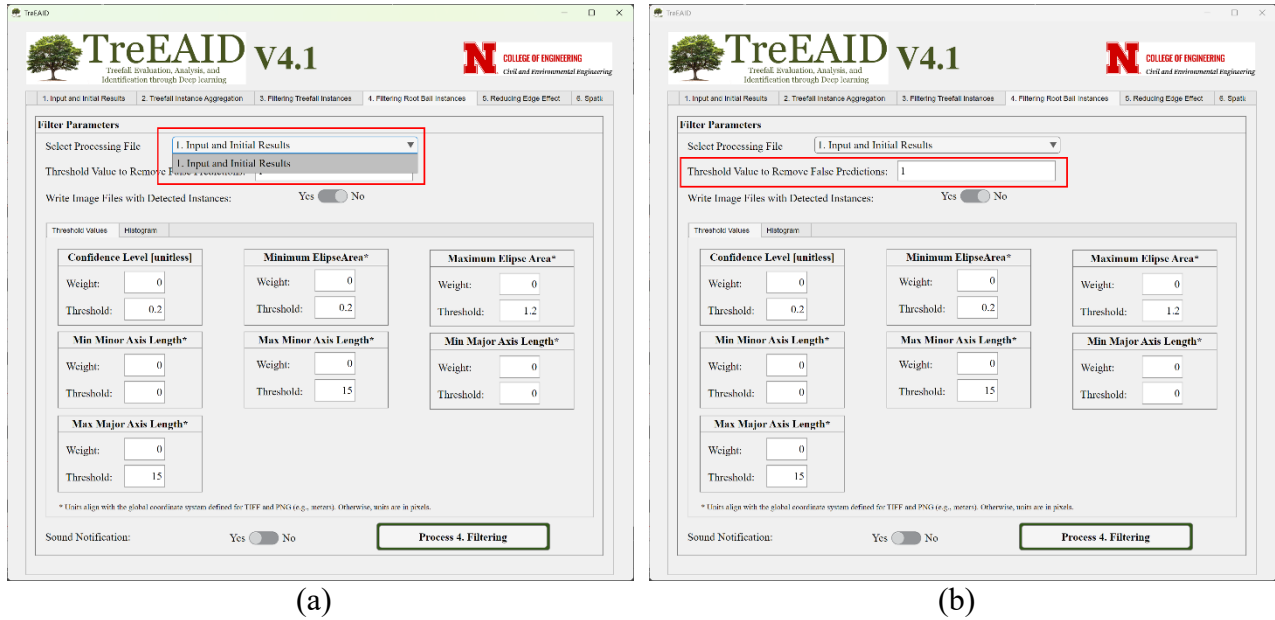


Figure 25. Filtering root ball instances: (a) selecting the dataset; (b) setting the threshold value to remove part of the data.

Various geometric parameters related to the physical features of the predicted root balls were evaluated during the parametric analysis. These filters help eliminate physically implausible or abnormally shaped detections. A combination of these parameters was found to effectively remove inaccurate predictions while retaining valid ones. It is important to note that the filtering parameters were established based on developer-analyzed datasets (with a GSD of approximately 2 cm). Users may need to optimize these parameters for their own dataset, depending on factors such as GSD, image quality, and tornado intensity. Table 8 summarizes the default threshold values for each filtering parameter. Users may also visualize the distribution of selected parameters, as demonstrated in Figure 26.

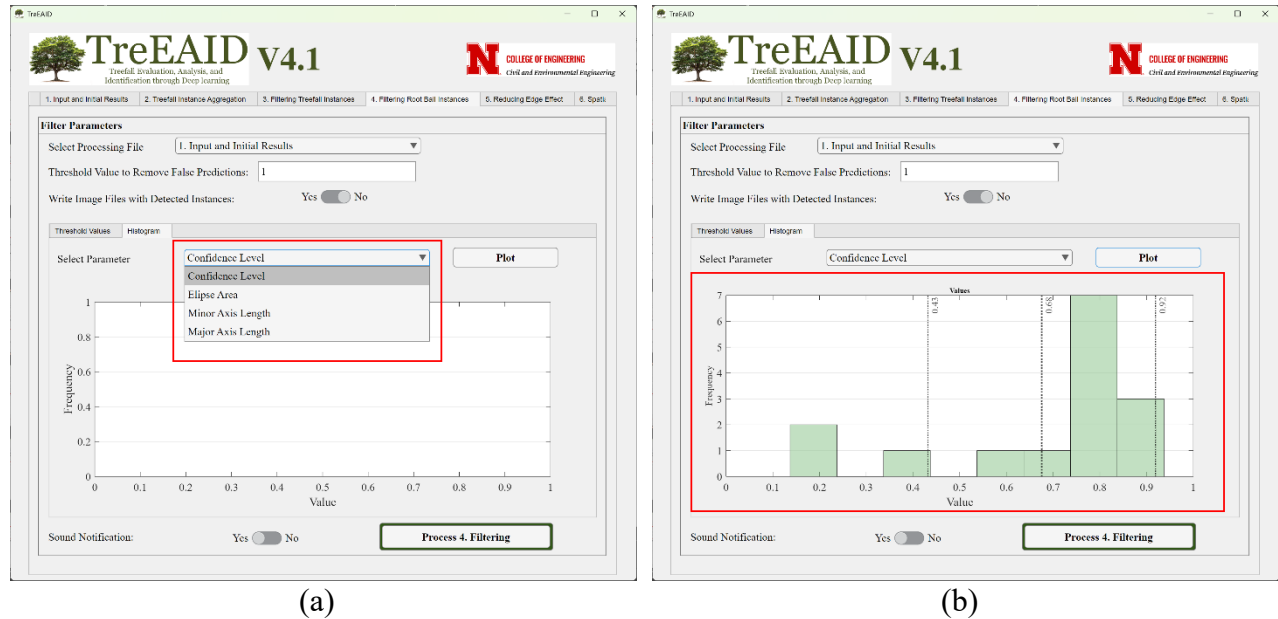


Figure 26. Visualization of root ball parameter distribution: (a) selecting parameters; (b) distribution of values with mean and standard deviation.

The filter parameters specified in this section include confidence level, minimum and maximum thresholds for ellipse area, minor axis length, and major axis length. Please note that the range of parameters for images in world coordinates and pixel coordinates varies significantly. The filtering parameters specified in this section can be defined as follows:

- Confidence Level: This parameter represents the probability that a prediction is correct. It is expressed as a decimal percentage.
- Minimum Ellipse Area: Represents the minimum area of the best-fit ellipse approximating the root ball. The unit is square meters (or square feet) for TIF and PNG images, and pixels for JPG images.
- Maximum Ellipse Area: Represents the maximum area of the best-fit ellipse approximating the root ball. Units follow the same convention, meters (or feet, if applicable) for TIF and PNG images, and pixels for JPG images.
- Minimum Minor Axis: Represents the minimum length of the minor axis of the best-fit ellipse. Units follow the same convention.
- Maximum Minor Axis: Represents the maximum length of the minor axis of the best-fit ellipse. Units follow the same convention.
- Minimum Major Axis: Represents the minimum length of the major axis of the best-fit ellipse. Units follow the same convention.
- Maximum Major Axis: Represents the maximum length of the major axis of the best-fit ellipse. Units follow the same convention.

Table 8. Statistical parameters are considered to filter root ball predictions.

Parameters	Symbol	Values	Units
Confidence Level	CL	< 0.2	%
Minimum Area	Min(A)		Global coordinate system (e.g., meters) or pixels (JPG)
Maximum Area	Max (A)		Global coordinate system (e.g., meters) or pixels (JPG)
Minimum Minor Axis	Min(b)		Global coordinate system (e.g., meters) or pixels (JPG)
Maximum Minor Axis	Max(b)		Global coordinate system (e.g., meters) or pixels (JPG)
Minimum Major Axis	Min(a)		Global coordinate system (e.g., meters) or pixels (JPG)
Maximum Major Axis	Max(a)		Global coordinate system (e.g., meters) or pixels (JPG)

11 Reducing Edge Effect

The "Reducing Edge Effect" tab is designed to mitigate issues associated with orthomosaic tiling artifacts. Specifically, this functionality addresses cases in which a single treefall instance is erroneously split across multiple image tiles, resulting in duplicated or fragmented detections. This tab enables the identification and merging of treefall segments that span adjacent tiles, reconstructing a single, continuous treefall vector representative of the entire fallen tree. This process is critical for applications that require one vector per physical treefall. To perform this task, the module compares adjacent tiles and uses the geographic alignment of treefall vectors to infer whether separate detections correspond to the same tree.

This step relies on precise geographic alignment and is only applicable to images in georeferenced formats, specifically TIF and PNG files associated with valid world files. JPG images lacking spatial metadata cannot be used in this stage. Furthermore, this process is applied exclusively to fallen trees; root balls are not considered. As shown in Figure 27, users are required to select the appropriate dataset to process. Available options include:

- 1. Input and Initial Results:
Processes the outputs generated in the “Input and Treefall Results” tab.
- 2. Tree Instance Aggregation (Weighted Average):
Processes the outputs generated in the “Treefall Instance Aggregation” tab using a weighted average approach. The result aggregates initial treefall polygons based on spatial proximity and defined conditions (as set in Tab 2). Treefall directions are estimated by computing a weighted average of individual vectors, with weights derived from parameters such as confidence level, taper rate, and vector length.
- 2. Tree Instance Aggregation and Results:
Processes outputs from the “Treefall Instance Aggregation” tab using reprocessed aggregated polygon masks. Instead of using weighted averages, the direction of each treefall instance is estimated directly from the taper rate of the final aggregated polygon.
- 3. Filtering Treefall Instances:
Processes the output generated from the “Filtering Treefall Instances” tab. If this dataset is selected, users must also specify the initial dataset that was used during filtering.



Figure 27. Defining the input dataset for reducing the edge effect

The parameters in this section are used to identify and group segments of the same tree located near image tile boundaries. Each parameter is defined below and illustrated in Figure 28:

- **Boundary Size [m]:** Defines the edge buffer region where tile boundaries are considered. Any treefall intersecting this region is evaluated for cross-tile merging.
- **Length of Extension ($L_{\text{extension}}$) [m]:** This parameter defines the outward extension of the region of interest (ROI) from each end of the fallen tree, as aligned with the tree. It accounts for gaps or discontinuities in the tree trunk that may result from occlusion, fragmentation, or reduced visibility in the image. The ROI enables the algorithm to assess whether disconnected segments may belong to the same tree. The length of this extension is user-defined and is applied symmetrically at both ends of the tree polygon.
- **Tree Thickness Multiplier (N) [unitless]:** This parameter is a user-defined scaling factor that determines the lateral width of the ROI used in the analysis. It scales the ROI width based on the predicted tree's thickness (t), defined as the median distance between the sides of the tree polygon. The ROI is centered on the treefall vector and extends symmetrically and perpendicularly as a function of N and t . For example, if $N = 2$, the ROI extends $2 \times t$ on each side, for a total lateral ROI width of $4 \times t$. This scaling keeps the ROI proportional to tree geometry, preventing it from being too narrow (missing nearby tile instance) or too wide (introducing irrelevant data, especially for cross-tile matches).
- **Tree Overlap Range (m):** This parameter is used to prevent the erroneous merging of parallel but distinct trees. For two segments that may belong to the same tree but are separated by a gap, their projections along the X and Y axes are evaluated. If the overlap between two tree prediction vectors exceeds a user-defined threshold in either direction, the segments are

considered separate trees—regardless of visual proximity—and are excluded from merging. This ensures that parallel, closely aligned trees are not mistakenly treated as parts of a single instance. This threshold is defined in the same units as tree length.

- Angle Difference (Degree): This threshold defines the maximum allowable angular deviation between two treefall segments being evaluated for aggregation. If the angle between the orientation vectors of two segments exceeds this threshold, they are treated as separate instances. This ensures that only tree segments with consistent fall directions are considered parts of the same fallen tree.

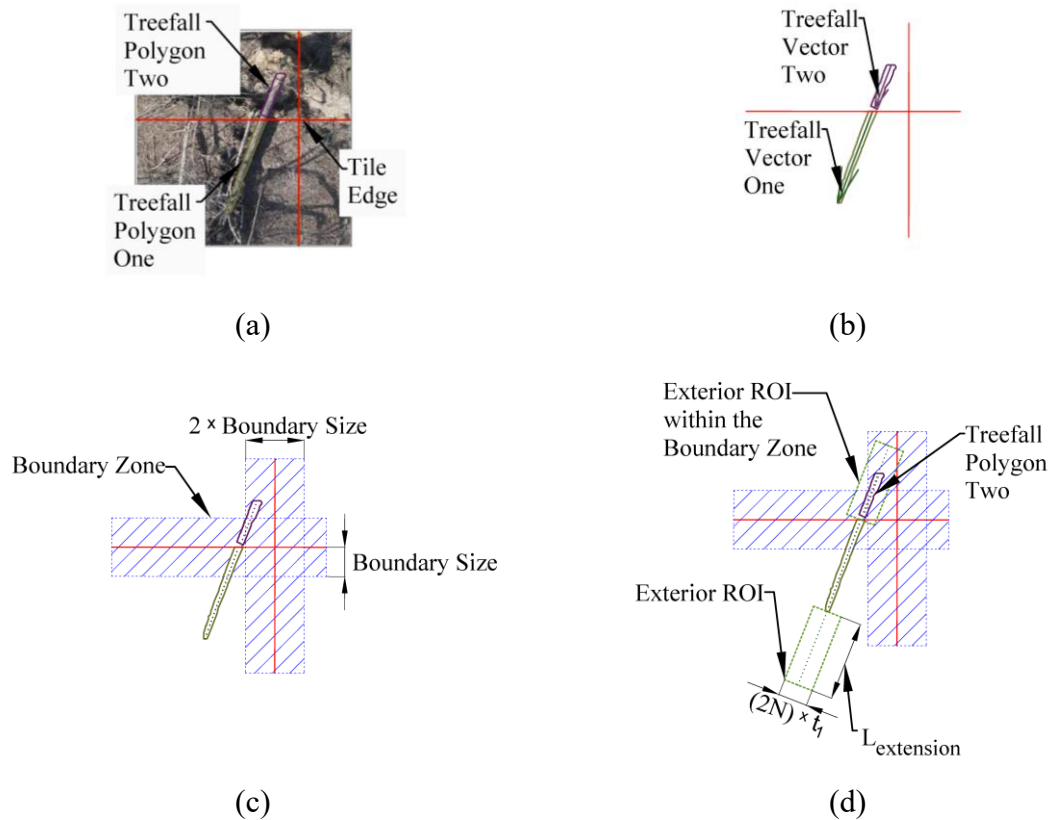


Figure 28. Parameters for reducing edge effect: (a) treefall polygons tiled across adjacent tiles; (b) estimated treefall vectors for each polygon on adjacent tiles; (c) boundary zone with specified boundary size; and (d) exterior ROI placement for treefall polygon one, with the critical ROI highlighted within the boundary zone for analysis.

When tiles are processed individually, the same fallen tree may be detected multiple times if it spans across adjacent images. This can lead to inflated treefall counts and inconsistencies in directional analysis. The *Reducing Edge Effect* step addresses this issue by identifying and merging treefall vectors near tile boundaries that likely represent the same event. As shown in Figure 29, this process significantly improves the accuracy of both the number and directional orientation of the resulting treefall vectors.

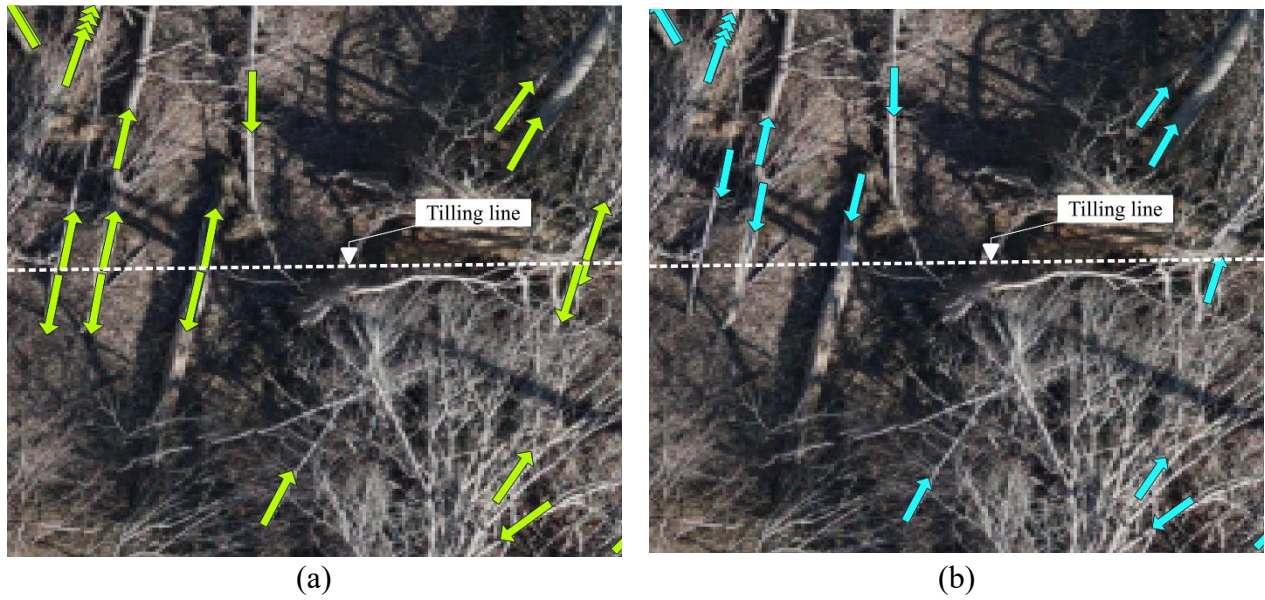


Figure 29. Reducing the edge effect of the prediction vectors shown in the QGIS: (a) tile-based vector outputs; (b) improved result after edge effect reduction.

12 Spatial Averaging of Treefall Instances

The “Spatial Averaging of Treefall Instances” tab enables users to subsample the dataset and estimate wind direction patterns across a region based on the distribution and orientation of treefall vectors. This approach is useful for generating a spatially averaged wind field from localized treefall information. To perform this operation, users must specify the following parameters:

- **Select Processing File:** The input file is selected from a dropdown menu. For example, the *Input and Initial Results* option corresponds to the dataset produced in Tab 1, whereas the *Reducing Edge Effect* option references data from Tab 5.
- **Grid Height and Width:** Defines the spatial resolution of the grid cells used for spatial averaging (e.g., $75 \text{ m} \times 75 \text{ m}$).
- **Minimum Tree Count:** Sets the minimum number of treefall vectors required within a grid cell for that cell to be included in the output. Cells that do not meet this minimum are excluded from the spatial averaging process.

Figure 30 displays the interface section where the processing file and spatial averaging parameters are specified by the user.

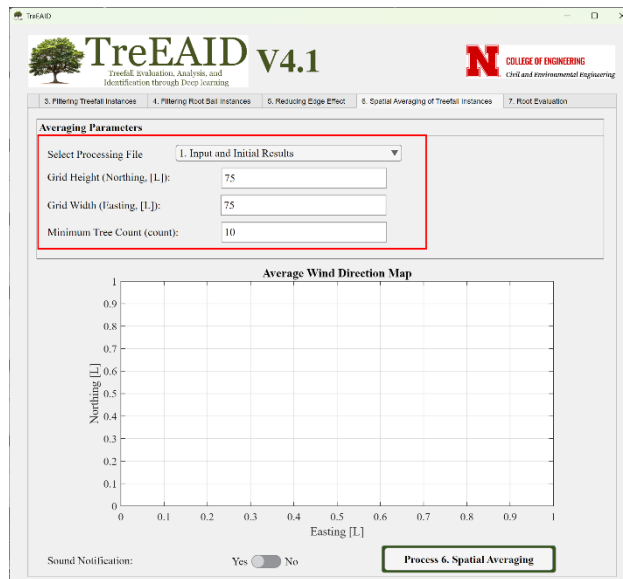


Figure 30. User interface section for defining the spatial averaging parameter.

The spatial averaging algorithm operates by calculating the median orientation of all treefall vectors within each grid cell that meets the minimum tree count. This vector is then placed at the center of the corresponding cell, representing the local average treefall direction. The user-defined grid size and tree count threshold should be selected based on contextual factors such as tornado intensity, path width, and vegetation structure (e.g., forest density and tree spacing). These inputs may require adjustment to account for the density of fallen trees and the size of the surveyed region. As illustrated in Figure 31, each arrow placed at the center of a grid cell represents the median direction of all treefall vectors within that cell. Small vectors indicate individual treefall orientations, while the larger arrows show spatially averaged directions across $75 \text{ m} \times 75 \text{ m}$ grid cells.

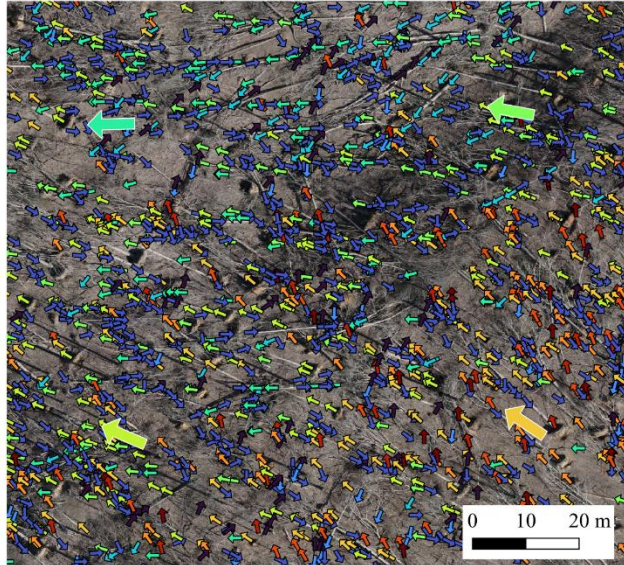


Figure 31. Spatial Averaging of treefall instance directions: small vectors represent each fallen tree orientation; large vectors represent median treefall direction per 75 m grid.

13 Root Evaluation

The “Root Evaluation” tab enables users to spatially aggregate root ball detections to estimate uprooting intensity across a surveyed region. This function computes both the average root ball area and total root ball count within a user-defined grid, providing insights into subsurface damage patterns caused by tornadoes or other disturbance events. To perform the analysis, the following user-defined inputs are required and are illustrated in Figure 32:

- **Select Processing File:** Choose the dataset to process from the dropdown menu. For example, *Input and Initial Results* refers to the raw root ball detection results generated in Tab 1, while *Filtering Root Ball Instances* refers to results refined through the filtering logic in Tab 4.
- **Grid Height and Width:** Define the horizontal dimensions of the spatial aggregation grid. These should reflect storm characteristics (e.g., tornado width and intensity), as well as local environmental, soil, and vegetation conditions.

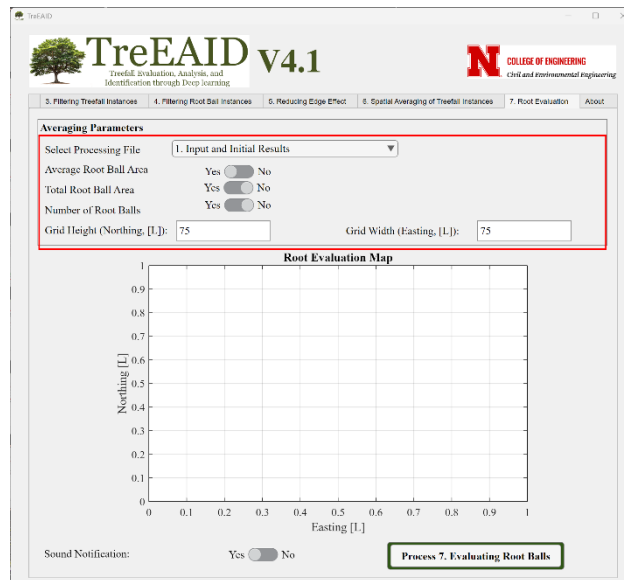


Figure 32. User interface section for defining root ball averaging parameters.

The algorithm processes the surveyed region to determine both the count and average area of uprooted root balls, using the grid dimensions specified by the user. The interface displays a single processing result at a time, either the number of root balls or their average area. However, the exported output files (in Excel and CSV formats) include both the spatially averaged count and average area for each grid cell. Figure 33 shows example results from the root ball evaluation module visualized in QGIS.

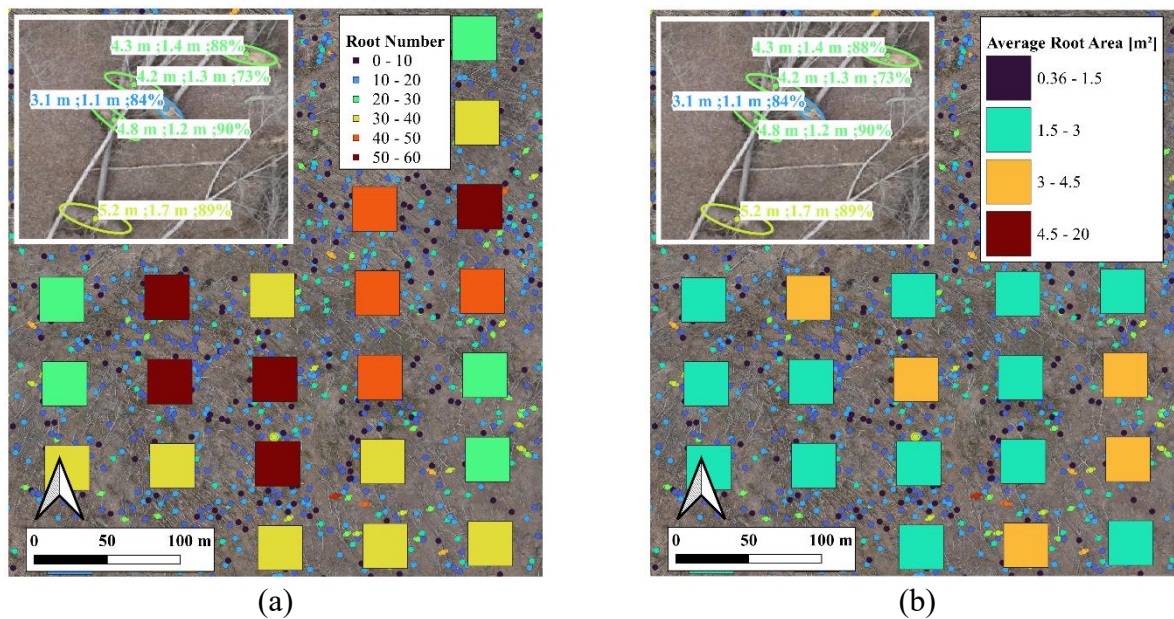


Figure 33. Root ball spatial averaging results in QGIS: (a) number of root balls; (b) average root ball area.

14 About

This application remains under active development, with continued efforts focused on enhancing its functionality, performance, and usability. Feedback from users is essential to this process. Users are strongly encouraged to share **comments, suggestions, or concerns** with the development team. All feedback is **greatly appreciated** and will be considered as part of the application's ongoing improvement process. By using the application, users acknowledge and accept the terms and conditions outlined in Figure 34. Please direct any feedback to:

- Prof. Richard L. Wood – rwood@unl.edu
- Mitra Nasimi – mnasimi2@huskers.unl.edu



Figure 34. User agreement and software use terms displayed within the application.

15 Reference

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Appendix A: Troubleshooting (Error Messages)

Here is a summary of error messages and guidance on resolving them:

1. Error: AttributeError: 'MexPrinter' object has no attribute 'encoding'.

This error indicates the lack of the required Python package. Users should ensure the installation of “Ultralytics,” “opencv-python,” and “roboflow” as detailed in the “Required Software and Packages” section. Figure 35 shows the error message indicating a missing supporting package, and Figure 36 shows how to confirm the Ultralytics installation from the command window.

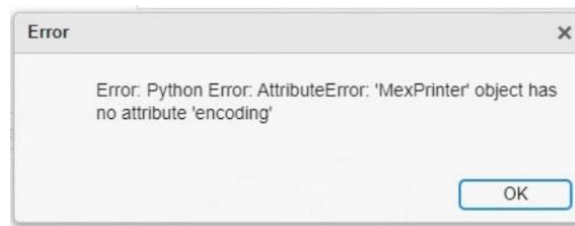


Figure 35. Lack of the required package.

```
Command Prompt
Microsoft Windows [Version 10.0.22631.3296]
(c) Microsoft Corporation. All rights reserved.

C:\Users\mhzna>pip show ultralytics
Name: ultralytics
Version: 8.0.208
Summary: Ultralytics YOLOv8 for SOTA object detection, multi-object tracking, instance segmentation, pose estimation and
image classification.
Home-page: https://github.com/ultralytics/ultralytics
Author: Ultralytics
Author-email: hello@ultralytics.com
License: AGPL-3.0
Location: c:\users\mhzna\appdata\roaming\python\python310\site-packages
Requires: matplotlib, numpy, opencv-python, pandas, pillow, psutil, py-cpuinfo, pyyaml, requests, scipy, seaborn, thop,
torch, torchvision, tqdm
Required-by: yolov5

C:\Users\mhzna>
```

Figure 36. Checking package installation using the command window.

2. Error: Python Error: AttributeError: Can't get attribute 'C3k2'.

This error indicates that the installed Ultralytics package on the system is outdated to the deep learning model loaded in the application. Updating Ultralytics will resolve the mismatch. Figure 37 shows the error message indicating a need to upgrade Ultralytics package, and Figure 38 shows how to upgrade the installation of the Ultralytics package using the command window.

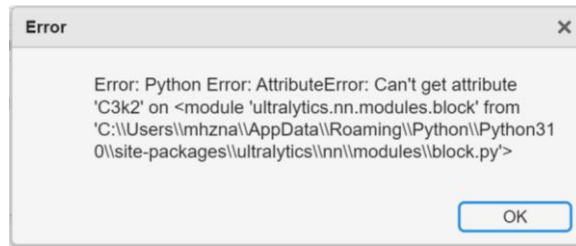


Figure 37. Outdated Ultralitics package error.

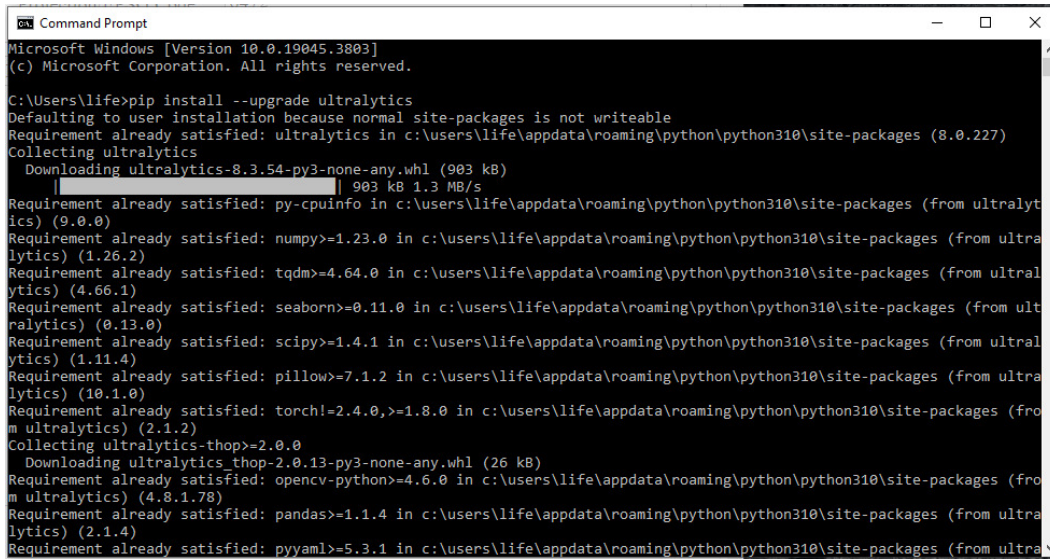


Figure 38. Upgrading the Ultralitics package using the command window.

3. Input Error: Please fill all required boxes.

This error may occur if one of the required input boxes is missing. Users should check the “Input and Treefall Results” tab to ensure that all required fields are filled correctly. Figure 39 shows the error notice shown due to missing input.

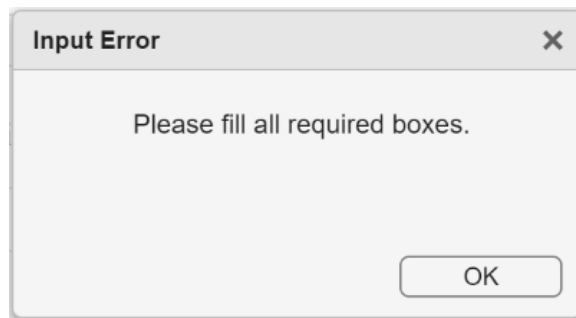


Figure 39. Missing input weight file error.

4. Input error: Please select weight file path.

This error occurs in the local model and indicates that the user did not select the weight file. To resolve this error, users must select the weight file provided with the application, as depicted in Figure 40.

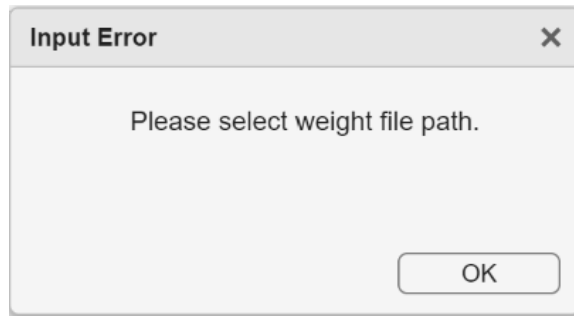


Figure 40. Missing input weight file error.

5. Error: Unrecognized function or variable 'new'.

This error can occur due to the selection of an inaccurate image file type/format or the absence of images in the selected format in the chosen folder. Users should verify the input folder's data type. Figure 41 demonstrates the error shown due to an inaccurate image format

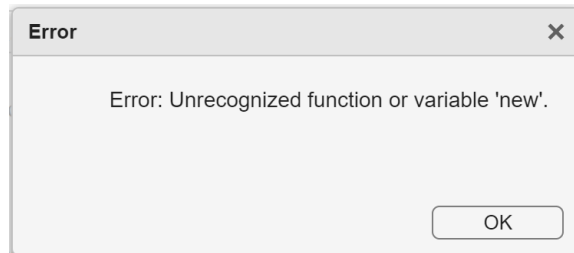


Figure 41. The error occurs when the specified file is not in the folder.

6. The file 'C:....pgw' could not be opened because: No such file or directory.

This error is associated with PNG images and occurs due to the absence of PGW files required to support PNG images in the input file. Users must check the selected folder for accuracy and ensure it includes scaling and orientation (PGW) information for all input images. Figure 42 shows the error due to the lack of world file for the PNG image format.

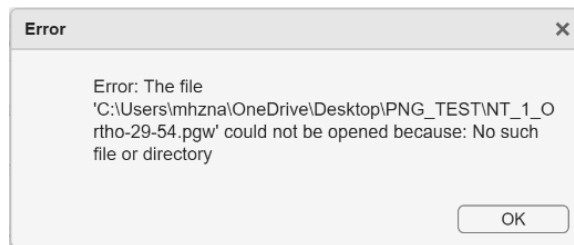


Figure 42. Lack of PNG and geographic information.

7. Error: Dot indexing is not supported for variables of this type.

This error is linked to the inaccurate selection of the EPSG code. Users should ensure the correctness of the EPSG code provided in the “Input and Treefall Results” tab. Figure 43 shows the error related to an invalid projection code.



Figure 43. Error caused by invalid EPSG code selection (dot indexing not supported).

8. Error: Argument must specify a projected coordinate reference system.

This error can occur due to the selection of an incorrect EPSG code. Users need to verify the provided EPSG code in the “Input and Treefall Results” tab. Figure 44 shows this error.

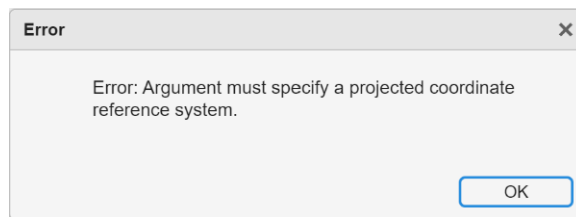


Figure 44. Error caused by invalid EPSG code selection (argument must specify a projected CRS).

9. Error: Index in position 2 exceeds array bounds.

This error can occur with PNG or TIF input data format, indicating that all grids have fewer trees than the specified minimum tree count threshold per cell. To resolve this, users can increase the grid size, or lower the minimum number of trees defined in the “Spatial Averaging of Treefall Instances” Tab. Figure 45 shows the error related to a high minimum tree number.

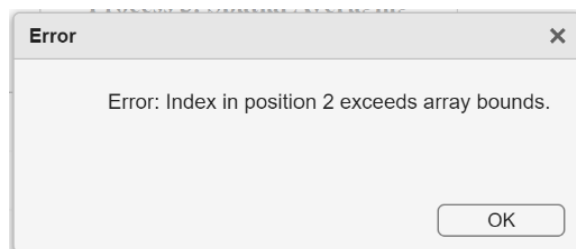


Figure 45. Minimum tree-count threshold set too high.

10. Error: Unrecognized function or variable ‘treefallTable’. Large file size.

This error occurs due to a large amount of input data or the detection of too many trees. Although the application can process large and small datasets, dividing the images into two or three folders may be necessary if this error occurs, as shown in Figure 46.

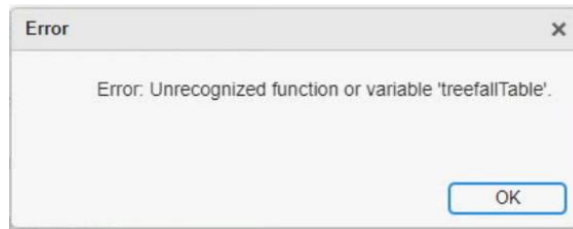


Figure 46. Large file size.

11. Error: csv could not be removed.

This error is caused by the files being opened in a different application, such as QGIS/ArcGIS. Users should ensure that all related documents are closed before reprocessing, as the application cannot rewrite the documents that are already open in other platforms. Figure 47 shows the error related to the inability to save a file when opening it in another program.

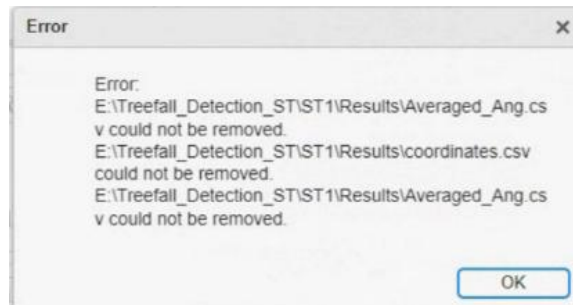


Figure 47. The file cannot be saved since it is open in another program.

12. Error: Python Error: Permission Error: [WinError 5] Access is denied

This error typically occurs when one of the files is open or due to insufficient admin access. Figure 48 shows the error related to the access problem. Users should ensure that:

- All related JSON files are closed before reprocessing the dataset.
- The input folder is located on a local drive (not on cloud storage servers such as OneDrive).
- If the issue persists, try running the application using the “Run as Administrator”

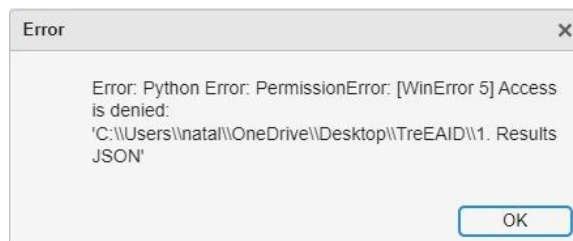


Figure 48. Permission access error.